Decision Analytics of Switchback Experiments: A Robust Optimization Approach

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Problem definition: Digital platforms often run switchback experiments, toggling the entire service between a new version (treatment) and the standard one (control) over time so users cannot influence each other, but analysts see only a few hundred periods instead of millions of users. Classical statistical inference—p-values and confidence intervals—quantifies uncertainty, but they give managers no direct sense of whether to roll out or shelve the change. This paper studies how to make the decision between treatment and control based on the observations a switchback experiment provides. **Methodology/results:** We consider a robust optimization approach that treats the outcomes that were not observed in each time period as unknowns and ask which choice—adopt or stay with the status quo—has the smaller regret under the worst-case scenario. By incorporating a Lipschitz continuity condition that links the outcomes of consecutive outcomes into the uncertainty set, we obtain the optimal robust solution that leverages the temporal structure inherent in switchback experiments. We extend the robust optimization approach to the settings with carryover effects and strengthen the uncertainty set by assuming the impact of carryover effect diminishes in time. Extensive simulations show that the proposed approach improves the expected regret while keeping the same safeguard against false alarms. **Managerial implications**: The robust optimization method requires no parametric or distributional assumptions, and delivers clearer, faster go/no-go answers than conventional significance tests. Practitioners can plug this into existing experimentation workflow as an early filter that drops weak ideas and frees scarce experimental resource for stronger ones. Researchers can build on the framework to handle other kinds of interference structure or pair it with classic tests in some other ways that both saves resources and boosts confidence in final launch decisions.

Key words: switchback experiments; robust optimization; causal inference; decision analysis

1. Introduction

Randomized controlled experiments form the cornerstone of experimental research across numerous disciplines, including biostatistics, medicine, and social sciences (Imbens and Rubin 2015). The standard workflow involves randomly assigning units—which could be individuals, groups, or time periods—to either a treatment group or a control group. This randomization aims to eliminate selection bias and ensure that any observed effects can be attributed to the treatment rather than confounding variables.

Once the data is gathered, statistical tools are employed to interpret the results and make informed decisions. A pivotal concept in this analysis is the Central Limit Theorem (CLT), from which confidence intervals can be calculated to estimate the range within which the true treatment effect lies with a specified level of confidence (e.g., 95%). Furthermore, statistical tests can determine whether the observed differences between the treatment and control groups are significant or could have occurred by random chance. For instance, a null hypothesis (no effect) is tested against an alternative hypothesis (there is an effect). The outcome of these statistical analyses informs the decision-making process. If the hypothesis testing indicates a statistically significant effect (p-value below a predetermined threshold like 0.05 or 0.1), and the confidence interval suggests a meaningful impact, the treatment may be adopted; if no significant effect is detected, or if the confidence interval includes negligible or negative effects, the status quo is typically maintained.

The use of the CLT and hypothesis testing offers two key advantages: (i) measurement of uncertainty: confidence intervals quantify the uncertainty around the estimated effects, offering insights into the precision and reliability of the results; (ii) control of Type I errors: hypothesis testing frameworks help manage the risks of false positives. Such analysis is universally applicable and is easy to implement in different areas: in biomedical research, clinical trials assessing new medications rely heavily on confidence intervals and hypothesis tests to determine efficacy and safety (Pocock 2013, Schulz et al. 2010); in political sciences, researchers employ statistical inference to evaluate the impact of policies, providing evidence-based recommendations (Cohen 1994, Gerber and Green 2012); in economics, hypothesis testing is applied to assess the effect of education on earnings and incomes (Angrist and Krueger 1991, Imbens and Wooldridge 2009).

In the past decades, experimentation has become a standard practice for evaluating the changes to products and services in business operations and innovations — particularly within tech companies (Thomke 2003, Kohavi and Thomke 2017, Cui et al. 2018, Gupta et al. 2019, Bojinov and Gupta 2022, Larsen et al. 2023). These changes are typically small and incremental, and the principle is to determine whether a proposed change (treatment) outperforms the current baseline (control), enabling rapid iteration and data-driven deployment. Despite the wide use of statistical tools in post-experiment analysis for decision making, modern experimentation faces several significant challenges.

First, a fundamental assumption of A/B testing is the non-interference between experimental units—that is, the treatment assigned to one unit does not affect another unit's outcome. In many experiments in online platforms and marketplaces, especially those involving network effects or shared environments, this assumption is violated. For example, changes to a ride-sharing app's pricing algorithm might influence market dynamics, affecting both treated and control users. To mitigate interference, companies often resort to switchback experiments (Bojinov and Shephard

2019, Bojinov et al. 2022, Glynn et al. 2020, Hu and Wager 2022, Xiong et al. 2022, Ni et al. 2023, Chen and Simchi-Levi 2023, Jia et al. 2024, Weng et al. 2024, Ni and Bojinov 2025), where the treatment assignment is randomized over time rather than across units. In a switchback design, the entire system switches between treatment and control conditions at different time periods. While this approach avoids interference, it typically leads to low statistical power. Consider an experiment where the system alternates treatments every hour over two weeks. This setup yields only 336 observations (24 hours $\times$ 14 days), a relatively small sample size compared to user-level experiments that can involve thousands or even millions of users. The limited number of time periods reduces the ability to detect small to moderate treatment effects, making it harder to draw conclusive results from the data.

Second, the power issue is even more substantial under experiment proliferation. Tech companies often face the need for running a huge number of experiments, far exceeding the volume in traditional medical or economic research settings. Google, Microsoft and LinkedIn each report conducting 20,000+ experiments annually (Kohavi et al. 2020), and LinkedIn alone juggles 400 concurrent tests every day (Xu et al. 2015). Larsen et al. (2023) observe that "industry giants run hundreds of experiments on millions of users every day". This proliferation is driven by rapid product iterations, algorithm developments, and a culture of data-driven decision-making. However, experimental resources such as time slots in switchback experiments are limited. The finite number of periods available for testing means that each experiment must compete for time, further diminishing the power per experiment. The scarcity of experimental resources intensifies the difficulty in detecting treatments with positive effects.

Third, in many corporate environments, the primary objective of experimentation is often to make optimal business decisions rather than to achieve precise estimates of treatment effects or strictly adhere to conventional statistical significance thresholds like p-values of 0.05 or 0.1. This nuance in focus suggests that the quality of decisions should be judged by: (i) the reward that could be achieved if the treatment were not adopted; (ii) the loss that could be avoided if the treatment were adopted. Several industry experts and researchers have highlighted such weighing between risks and opportunities. Kohavi et al. (2009) discuss that in large-scale online experiments, the focus should be on the expected impact on key business metrics such as revenue and customer engagement, rather than solely on statistical significance. They emphasize that over-reliance on p-values can lead to missed opportunities or the implementation of changes that have negligible practical benefits. Similarly, Kohavi et al. (2020) argue that organizations should adopt a decision-making framework that balances statistical evidence with business considerations. They advocate for evaluating the magnitude of the effect, implementation costs, and how the change aligns with strategic objectives. These perspectives underscore the need for a holistic approach to decision-making in business

experimentation – companies must consider the expected impact on business metrics, the risks involved, and the opportunities presented.

1.1. Main Results

In this paper, we study the decision analytics of switchback experiments. We particularly ask the following question: given the observations collected from a switchback experiment, how can we make the decision choice between treatment and control?

We address this question through a robust optimization framework that delivers a minimax optimal decision rule. Because a switchback design reveals only one potential outcome per period, the other outcome remains unobserved, leading to the uncertainty of the problem. We evaluate each decision by its worst-case simple regret, computed under an adversary that assigns the missing outcomes. The choice with the smaller worst-case regret is then selected. To formulate the robust optimization, we consider two different uncertainty sets with various constraints on the unobserved outcomes, both of which lead to explicit optimal decision rules without the need of solving the optimization problems. The first uncertainty set constrains each missing outcome to be bounded. Under this specification, the optimal decision rule simplifies to checking the classical inverse-propensity-weighted (IPW) estimator commonly used in statistical inference. The second uncertainty set imposes a Lipschitz-continuity condition, linking nearby periods through a smoothness constraint. Its optimal solution is equivalent to constructing a new estimator that extrapolates the unobserved outcomes from local observations.

We analyze the statistical properties of the two robust optimization solutions and derive an explicit expression for the difference in their variances. By exploiting the outcome structure inherent to switchback experiments, we demonstrate that incorporating the Lipschitz-continuity improves performance. These results underscore the value of contextual structure in experimental analysis without imposing specific parametric model assumptions.

We extend the robust framework to settings with carryover effects. When carryover is present, each period's outcome depends not only on the current assignment but also on past assignments, creating an expanded set of potential outcomes indexed by assignment history. We adapt our Lipschitz-based uncertainty set to this enlarged space. More importantly, we construct a more informative uncertainty set that connects the potential outcomes under different assignment scenarios by assuming that the impact of the carryover effect diminishes in time. The resulting robust problem can be reformulated as a mixed-integer linear program that is computationally tractable.

We conduct a comprehensive numerical study that compares the robust optimization approach with classical hypothesis testing procedures. Across a wide range of scenarios, the robust decision rule yields smaller expected regret than hypothesis testing, even though it is derived from a minimax perspective. After calibrating its risk parameter in the objective function, the robust decision rule

also achieves a lower Type II error while maintaining the same Type I error rate as the classical hypothesis testing.

We position robust optimization as a complement rather than a replacement for classical statistical inference methods. In the numerical study, we also consider a resource-constrained environment with many concurrent experiments and propose a two-stage strategy: in Stage 1, apply the robust optimization to screen out experiments that may be harmful; in Stage 2, subject the survivors to conventional hypothesis tests to obtain formal statistical guarantees. Simulations show that this hybrid approach substantially increases statistical power without sacrificing Type I control, underscoring the value of robust optimization as an effective screening phase in modern experimentation.

1.2. Related Works

Experimental practice typically encompasses three key steps: design, estimation, and decision-making. While a vast body of literature explores each of these components using different advanced techniques—including emerging work on machine learning and generative AI—the focus here is narrower. We concentrate on studies that leverage modeling or decision-driven tools across these steps, offering a distinct perspective on the analytics of experiments.

The design of experiments is a systematic approach to implementing experiments that optimizes data efficiency while minimizing cost and effort. It originated with the pioneer's work Fisher (1936), which established key principles such as randomization, replication, and blocking to ensure reliable inference, followed by works (Wu 1981, Bickel and Doksum 2015) that analyze experiments from a decision analysis perspective. In recent decades, the scope of experimentation has broadened significantly as new challenges have emerged in dynamic and complex environments. Recent business research—especially in operations and marketing—is increasingly embracing experimental design to tackle the inherent complexity of business contexts (Bojinov et al. 2022, Ni et al. 2023, Xiong et al. 2024, Holtz et al. 2020). Traditional designs, developed for more controlled industrial and agricultural settings, are often insufficient when human behavior, dynamic market conditions, and social interactions come into play. For example, the presence of interference—where the effect of a treatment on one unit spills over to influence others—complicates the clear delineation of direct and indirect effects, necessitating innovative designs that explicitly account for network structures and spillover effects. The high-level idea is to formulate the design of experiments as an optimization problem and study the optimal design with different objective functions. We refer to Zhao (2024) for a comprehensive review with more examples.

The analysis of experiments has increasingly focused on developing efficient estimators that accurately capture causal effects, especially in complex environments where traditional estimators fall short. Similar to experimental design, estimation precision can be severely compromised by

interference, which may introduce bias or even reverse the sign of the true causal effect. To address these challenges, many studies leverage the inherent structure of the problem—exploiting specific model characteristics to reduce bias or decrease variance—with applications spanning inventory optimization, queuing systems, one-sided/two-sided markets, general Markov dynamic processes, and random graphs (Si 2023, Zhu et al. 2024, Hu and Wager 2022, Jia et al. 2024, Farias et al. 2022, Li et al. 2022, Zhan et al. 2024, Chen et al. 2024, Li et al. 2023, Shirani and Bayati 2024, Dhaouadi et al. 2023). Notably, Bright et al. (2022) examines bias reduction in marketplace experiments, where interference arises from solving a linear optimization as a matching algorithm.

The decision-making of treatment choice based on experimental data is often studied with the objective of maximizing welfare. Similar to our work, a few studies have explicitly employed a minimax optimization framework to determine treatment choice (Manski 2007, Stoye 2012) from a sampling-based perspective, though not within the context of switchback experiments. In contrast, much of the literature centers on individualized treatment rules—a framework also referred to as policy learning (Kitagawa and Tetenov 2018, Kallus and Zhou 2021, Zhang et al. 2024). In this approach, researchers construct specific objective functions that capture the trade-offs between different candidate treatment policies and then solve the resulting optimization problems to derive minimax optimal solutions, ranging from simple linear decision rules to more complex machine learning models such as random forests. Given that the focus of this paper is on a straightforward treatment/control decision that leverages the problem's structure, we refer readers to Wager (2024) for a detailed discussion and examples on policy learning. Moreover, Johari et al. (2024) examine decision-making in experiments on online platforms and marketplaces. Their study analyzes how interference impacts decision-making in a queuing model, but they still rely on a frequentist hypothesis testing framework.

2. Setup and Notations

Suppose that there are T time periods. At each time point $t \in [T] = \{1, 2, \ldots, T\}$, we assign the unit to receive an intervention $W_t \in \{0, 1\}$. For example, one experimental period could be one hour and T could be two weeks, i.e., $T = 336$ when one period is one hour. Following convention, we say that the unit is assigned to treatment if $W_t = 1$ and control when $W_t = 0$. The assignment path is then the collection of assignments and is denoted using a vector notation whose dimensions are specified in the subscript, $\boldsymbol{W}_{1:T} = (W_1, W_2, \ldots, W_T) \in \{0, 1\}^T$. We adopt the convention that $\boldsymbol{W}_{1:T}$ stands for a random assignment path, while $\boldsymbol{w}_{1:T}$ stands for one realization.

Following the extended potential outcomes framework, at time $t \in [T]$, we posit that for each possible assignment path $\boldsymbol{w}_{1:T}$ there exists a corresponding potential outcome denoted by $Y_t(\boldsymbol{w}_{1:T})$. For example, this could refer to any business metric during one experimental period. The set of

all potential outcomes is collected in $\mathbb{Y} = \{Y_t(\boldsymbol{w}_{1:T})\}_{t\in[T],\boldsymbol{w}_{1:T}\in\{0,1\}^T}$ with support $\mathbb{Y} \in \mathcal{Y}$. In this paper, we adopt a design-based perspective (Neyman 1923, Fisher 1936, Kempthorne 1955, Rubin 1980, Imbens and Rubin 2015, Abadie et al. 2020) and treat the potential outcomes $\mathbb{Y}$ as fixed but unknown quantities, or equivalently, we condition on $\mathbb{Y}$. The design-based approach does not impose strong modeling assumptions on outcomes, offering robustness to complex dynamics and interference patterns common in operational settings. We are interested in the global average treatment effect

$$\tau = \frac{1}{T}\sum_{t=1}^{T}(Y_t(\mathbf{1}) - Y_t(\mathbf{0})) \tag{1}$$

which measures the average difference in outcomes that would be observed if all periods are treated (denoted by $\mathbf{1}$) compared to if no period is treated (denoted by $\mathbf{0}$).

In this paper, we consider the completely randomized design of experiments (Neyman 1923, Li and Ding 2017, Ni et al. 2023) where half of the periods are assigned to the treatment group and the other half are assigned to the control group. After an experiment is done, based on the observed assignment path $\boldsymbol{W}_{1:T}^{\text{obs}}$, we collect the observed potential outcomes $Y_t^{\text{obs}} = Y_t(\boldsymbol{W}_{1:T}^{\text{obs}}), \forall t \in [T]$, from which we then make the decision between adopting the treatment ($Z = 1$) or keeping the control ($Z = 0$). We use the notion of simple regret (Bell 1982) to evaluate the quality of a decision, which is given by

$$\mathcal{R}(\mathbb{Y}, Z) = \frac{1}{T}\sum_{t=1}^{T}(Y_t(\boldsymbol{Z}^\tau) - Y_t(\boldsymbol{Z})) \tag{2}$$

where $Z^\tau = \mathbb{1}\{\tau > 0\}$, $\boldsymbol{Z} = Z \cdot \mathbf{1}$ and $\boldsymbol{Z}^\tau = Z^\tau \cdot \mathbf{1}$.

The simple regret quantifies the average "cost" across all periods of not choosing the best possible decision based on the outcomes we would have observed under treatment and control. This provides a clear metric for evaluating how much benefit is lost due to the uncertainty in the decision-making process, which is widely adopted for the treatment choice with empirical welfare maximization (Kitagawa and Tetenov 2018, Athey and Wager 2021). By focusing on simple regret, we align our evaluation metric with the practical concern of ensuring that the treatment decision minimizes the loss relative to the best possible outcome that could have been achieved with perfect information.

3. Robust Optimization

The objective of our decision analytics of a switchback experiment is to choose $Z \in \{0, 1\}$ that minimizes the simple regret $\mathcal{R}(\mathbb{Y}, Z)$. Unfortunately, we never observe the entire set of outcomes $\mathbb{Y}$, but only a subset of observed outcomes, leading to a typical setting of decision making under uncertainty. To tackle this, robust optimization (Bertsimas and Sim 2004, Ben-Tal et al. 2009) becomes an effective and practical approach, where one specifies an uncertainty set—a range or collection of plausible values that uncertain parameters (i.e. unobserved potential outcomes) might

take—and then optimizes the decision based on the worst-case scenario within that set. The goal is to "immunize" the decision against adverse outcomes that would be observed. Precisely, we have the following problem formulation:

$$\min_{Z\in\{0,1\}} \max_{\mathbb{Y}\in\mathcal{Y}^{\text{obs}}} \mathcal{R}(\mathbb{Y}, Z) \tag{3}$$

where $\mathcal{Y}^{\text{obs}} \subset \mathcal{Y}$ is the uncertainty set constructed from the observed outcomes. It defines the domain of potential outcomes that the adversary is allowed to select from, based on what has been revealed in the experiment. The use of minimax regret in decision analytics is well-established, supported by the literature in studying treatment rules and sample selections (Manski 2007, Stoye 2012, Kitagawa and Tetenov 2018, Kallus and Zhou 2021, Hu et al. 2024). Our approach differs importantly from the traditional sampling-based methods common in existing literature. Instead of relying on probabilistic assumptions or sampling distributions to construct uncertainty sets, our framework is distinctly design-based. This means we construct uncertainty sets directly from observed outcomes resulting from treatment assignments, adhering strictly to the randomization design of our experiment. Such a design-based approach avoids assumptions regarding distributional forms, thus providing a rigorous and assumption-light methodology particularly suitable for experimental environments like switchback designs.

Before proceeding to study the formulation, we introduce the following non-interference assumption that simplifies the analysis of switchback experiments.

Assumption 1 **(No temporal interference)**. *There is no temporal interference or carryover effects. That is,*

$$Y_t(\boldsymbol{w}_{1:T}) = Y_t(\boldsymbol{w}'_{1:T}) \text{ if } w_t = w'_t, \forall t \in [T]. \tag{4}$$

This is essentially the SUTVA in causal inference literature. In the context of switchback experiments, it means the potential outcome at period t only depends on the treatment assignment at current period t. There is a growing body of literature studying switchback experiments in the presence of carryover effect – past treatment may impact future outcomes. We will relax this assumption later in Section 5 to see how the robust optimization framework can be adapted to more complex settings.

3.1. Naive Uncertainty Set

To implement the robust formulation (3), we start with a simple uncertainty set $\mathcal{Y}_0^{\text{obs}}$, based on the following assumption.

Assumption 2 **(Bounded support)**. *The potential outcomes are bounded. That is*

$$\mathcal{Y} = \{Y_t(w_t) \in [0, C], \forall t \in [T]\}. \tag{5}$$

The necessity of this assumption is to avoid the case where the worst-case simple regret becomes infinitely large because one can always push the unobserved potential outcomes to either arbitrarily large or small if the support is not bounded. We do not necessarily require non-negativity, but this is a common scenario in practice, referring to most business metrics such as revenue, demand, click/purchase rate, delivery/travel time, etc.

Because of Assumption 1 and 2, there are two bounded potential outcomes at each period. We define two decision variables $y_t(1)$ and $y_t(0)$[1], and for any $y_t(w) \in \mathcal{Y}_0^{\text{obs}}$, we have

$$y_t(w) = Y_t^{\text{obs}}, W_t^{\text{obs}} = w;\ y_t(w) \in [0, C], W_t^{\text{obs}} \neq w. \tag{6}$$

We analyze the robust optimization under this uncertainty set. Suppose that the decision is $Z = 0$, then the worst-case regret must be attained at $Z^\tau = 1$. To see this, it is easy to check that the regret is zero at $Z^\tau = 0$ for any $\mathbb{Y}$. Let us consider the following feasible solutions: $y_t(1) = y_t(0), \forall t \in [T]$, under which $Z^\tau = 1$ also gives zero regret. If there exists $y_{t'}(0) < C$ for some t', we let $y_{t'}(1) = y_{t'}(0) + \epsilon$ and then the regret is ϵ at $Z^\tau = 1$. Therefore, $Z^\tau = 1$ is a dominant solution for maximizing worst-case regret. A symmetric argument holds for $Z = 1$. As a result, the robust optimization can be reformulated as

$$\min_{Z \in \{0,1\}} \max_{\boldsymbol{y} \in \mathcal{Y}_0^{\text{obs}}} \frac{1}{T} \sum_{t=1}^{T} \left(y_t(1 - Z) - y_t(Z) \right). \tag{7}$$

THEOREM 1. *The optimal solution of* (7) *satisfies*

$$Z^* = \begin{cases} 1, & \widehat{\tau} > 0 \\ 0, & \widehat{\tau} \leq 0 \end{cases} \tag{8}$$

where $\widehat{\tau}$ *is the standard IPW estimator given by*

$$\widehat{\tau} = \frac{1}{T} \sum_{t=1}^{T} Y_t^{\text{obs}} \left(\frac{\mathbb{1}\{W_t = 1\}}{\mathbb{P}(W_t = 1)} - \frac{\mathbb{1}\{W_t = 0\}}{\mathbb{P}(W_t = 0)} \right). \tag{9}$$

We prove this in Appendix A. The intuition is straightforward. Consider choosing to maintain the control: in the worst-case scenario, unobserved treatment outcomes would reach the upper bound while unobserved control outcomes would hit the lower bound, thereby maximizing the projected average treatment effect in favor of treatment. A symmetric argument applies if the decision were to adopt the treatment. Because the experiment allocates an equal number of treatment and control periods, these worst-case deviations cancel out, leaving the comparison of observed outcome averages as the decisive factor.

Under the formulation (7), we show that the optimal decision rule reduces to checking the inverse probability weighting (IPW) estimator that is independent of C—a statistical solution that mirrors

[1] Throughout the paper, we let Y_t be the true potential outcomes and use y_t to denote our decision variables for the optimization problems.

what one would obtain in typical analysis of A/B tests. This result arises because the naive uncertainty set does not capture any special structure unique to switchback experiments. Unlike individual-level experiments, experiments over time are inherently sequential: time has a natural order (e.g., 1 pm is followed by 2 pm and then by 3 pm). This ordering is fixed and cannot be arbitrarily permuted or relabeled. By not leveraging this temporal structure, the uncertainty set $\mathcal{Y}_0^{\text{obs}}$ ignores valuable information embedded in the time series, potentially limiting the power of the robust optimization.

3.2. Refined Uncertainty Set

Given the observations above, our goal is to refine the uncertainty set by considering the structure of switchback experiments. By doing so, we can capture sequential dependencies to enhance the reliability of the optimal robust solution, leading to a more nuanced and effective decision process that goes beyond the capabilities of the standard IPW estimator used in conventional A/B testing.

Specifically, we introduce Lipschitz continuity in the discrete setting. We impose the condition that the change in the outcome of interest between any two consecutive periods is bounded.

ASSUMPTION 3 **(Lipschitz continuity)**. *There exists a constant L that, for every adjacent pair of periods t and $t+1$, we have*

$$|Y_{t+1}(w) - Y_t(w)| \le L. \tag{10}$$

This constraint leverages the natural ordering of time in switchback experiments by ensuring that outcomes do not exhibit unrealistic jumps between consecutive periods. Incorporating this Lipschitz continuity, we introduce the refined uncertainty set $\mathcal{Y}_1^{\text{obs}}$ such that for any $y_t(w) \in \mathcal{Y}_1^{\text{obs}}$, we have

$$y_t(w) = Y_t^{\text{obs}}, W_t^{\text{obs}} = w;\ |y_{t+1}(w) - y_t(w)| \le L, W_t^{\text{obs}} \ne w, t < T. \tag{11}$$

Then we aim to solve the robust optimization under $\mathcal{Y}_1^{\text{obs}}$:

$$\min_{Z\in\{0,1\}} \max_{\mathbb{Y}\in\mathcal{Y}_1^{\text{obs}}} \mathcal{R}(\mathbb{Y}, Z). \tag{12}$$

Different from the naive uncertainty set, the formulation (12) does not lead to a simple closed-form optimal solution, but it can be solved with two linear programs. We let $\mathcal{R}(\mathbb{Y}^*(1), 1)$ denote the worst-case regret for $Z = 1$ and $\mathcal{R}(\mathbb{Y}^*(0), 0)$ denote the worst-case regret for $Z = 0$. To solve $\mathcal{R}(\mathbb{Y}^*(1), 1)$, we have

$$\begin{aligned}
\mathcal{R}(\mathbb{Y}^*(1), 1) = \max_{\mathbb{Y}\in\mathcal{Y}_1^{\text{obs}}} \mathcal{R}(\mathbb{Y}, 1) = \max \quad & \frac{1}{T}\sum_{t=1}^{T} (y_t(0) - y_t(1)) \\
\text{s.t.} \quad & y_t(W_t^{\text{obs}}) = Y_t^{\text{obs}}, \quad \forall t \in [T] \\
& -L \le y_t(0) - y_{t+1}(0) \le L, \quad \forall t \in [T-1], W_t^{\text{obs}} \ne W_{t+1}^{\text{obs}} \\
& -L \le y_t(1) - y_{t+1}(1) \le L, \quad \forall t \in [T-1], W_t^{\text{obs}} \ne W_{t+1}^{\text{obs}}
\end{aligned}$$

To solve $\mathcal{R}(\mathbb{Y}^*(0), 0)$, we keep the same feasible domain but change the objective to $\frac{1}{T}\sum_{t=1}^{T}(y_t(1) - y_t(0))$. Then it suffices to compare $\mathcal{R}(\mathbb{Y}^*(1), 1)$ and $\mathcal{R}(\mathbb{Y}^*(0), 0)$ to choose the action with a lower value. For implementation, we need to determine the parameter L that defines the constraints of our uncertainty set. A practical approach is to estimate it using historical data, particularly if we assume the treatment does not significantly alter the underlying smoothness characteristics of the data generation process. Fortunately, we demonstrate that the worst-case outcomes can be explicitly characterized in a way that is independent of L.

We first define a new estimator based on the observed outcomes. For the treatment group, let $\mathcal{T}(1)$ be the list of ordered indices where $W_t^{\text{obs}} = 1$ (e.g., $\mathcal{T}(1) = [3, 4, 6, 7, 10, 12, \ldots]$). For $t \in \mathcal{T}(1)$, we impute the missing treatment outcomes in the following way: for periods $\mathcal{T}_i(1) = r < t < s = \mathcal{T}_{i+1}(1)$, we let

$$Y_t^{\dagger}(1) = \frac{t-r}{s-r} Y_r^{\text{obs}}(1) + \frac{s-t}{s-r} Y_s^{\text{obs}}(1). \tag{13}$$

On the boundary cases, for any period $t < s = \min \mathcal{T}(1)$, we let $Y_t^{\dagger}(1) = Y_s^{\text{obs}}(1)$; for any periods $t > r = \max \mathcal{T}(1)$, we let $Y_t^{\dagger}(1) = Y_r^{\text{obs}}(1)$. We apply the same procedure to the control group and construct the locally-weighted estimator

$$\widehat{\tau}^* = \frac{1}{T}\sum_{t=1}^{T} \mathbb{1}\{W_t = 0\}\left(Y_t^{\dagger}(1) - Y_t(0)\right) + \mathbb{1}\{W_t = 1\}\left(Y_t(1) - Y_t^{\dagger}(0)\right). \tag{14}$$

This leads to the optimal decision rule for the robust optimization (12).

THEOREM 2. *The optimal solution of* (12) *satisfies*

$$Z^* = \begin{cases} 1, & \widehat{\tau}^* > 0 \\ 0, & \widehat{\tau}^* \leq 0 \end{cases} \tag{15}$$

We prove this in Appendix B. This characterization shows that the optimal robust decision can be determined without solving linear programs. Nonetheless, we begin by formulating linear programs because they lay the groundwork for more complex scenarios—such as when there are carryover effects (Section 5)—where the optimal decision rules are not readily available.

Typically, the primary goal in causal estimators is to infer unobserved outcomes from observed outcomes, either using model-based or model-free approaches. The standard IPW estimator (9) is equivalent to a uniform extrapolation where the unobserved outcomes are imputed from any permutation of all the observed outcomes. For switchback experiments, however, the refined uncertainty set explicitly incorporates the experiment's inherent structure, resulting in an optimal decision rule that leverages a locally-weighted estimator (14), where each unobserved outcome is imputed using linear extrapolation from its nearest observed outcomes. This method is intuitive because, for instance, if the outcome at today's 2 pm is missing, it makes sense to extrapolate using data from close periods, such as today's 1 pm and 3 pm, rather than data from random distant time points.

3.3. Comparison of Two Formulations

Theorem 1 and 2 not only provide explicit optimal decision rules from two robust formulations, but also, more importantly, establish a connection between the robust optimization solutions and the sample statistics, enabling more sophisticated analysis and comparison between the two robust formulations.

The primary goal of developing a robust optimization framework is not statistical inference per se; but investigating the statistical properties of the resulting solutions helps clarify how effective the decision rules are under different robust formulations. Although the properties of the IPW estimator, $\widehat{\tau}$, are well established, those of $\widehat{\tau}^*$ are far less straightforward. This complexity arises because the procedure to construct $\widehat{\tau}^*$ introduces nonregular correlations across time periods—the imputed outcome at any given period is randomly influenced by the outcomes of its neighbors. Analyzing this structure requires a sophisticated approach that characterizes conditional probabilities for every pair of periods, driven by the random permutation of assignments. In light of this complexity, we present our analytical results under the following two conditions:

- We consider the asymptotic regime as $T \to \infty$. Although the proof initially derives non-asymptotic results—entailing complex combinatorial sums and hypergeometric notations—we focus on the asymptotic case to extract the primary analytical insights.
- We introduce an additional constraint to the robust optimization framework defined in (12) by enforcing $|Y_1(w) - Y_T(w)| \leq L$. This measure mitigates boundary issues and ensures that all time periods exhibit a homogeneous covariance structure. Without this constraint, periods would have varying probabilities of extrapolation depending on their proximity to the boundaries (i.e., the first or last period). Although this additional constraint is superfluous, its effect on the imputed outcomes is very limited in the asymptotic regime of T.

We first investigate the expectations of the solutions. It is well known that the IPW estimator $\widehat{\tau}$ is unbiased. We further establish that the estimator $\widehat{\tau}^*$, derived from the optimal robust optimization problem in (7), is also unbiased:

PROPOSITION 1. $\mathbb{E}[\widehat{\tau}] = \mathbb{E}[\widehat{\tau}^*] = \tau$.

The proof is provided in Appendix C. From Theorems 1 and 2, the accuracy of a decision when the true treatment effect is positive is measured by $\mathbb{P}(\widehat{\tau} > 0 | \tau > 0)$ and $\mathbb{P}(\widehat{\tau}^* > 0 | \tau > 0)$. Since both estimators are unbiased, a natural quantity to compare is their variance, which—while not the sole determinant of precision—plays a critical role in shaping the distribution of the estimator and thus affects the probability of making correct decisions. Specifically, lower variance implies less dispersion, reducing the chance that noise alone flips the sign of the estimated effect. We show that the variance difference between $\widehat{\tau}$ and $\widehat{\tau}^*$ can be decomposed into a sum of cross–second moments of outcomes

separated by q periods, illuminating how temporal correlations influence the stability and reliability of each estimator.

PROPOSITION 2. *As $T \to \infty$, the difference of variance*[2] *is*

$$\begin{aligned} T^2\left(\mathrm{Var}(\widehat{\tau}^*) - \mathrm{Var}(\widehat{\tau})\right) = &\sum_{q=0}^{T/2} \beta_q^+ \sum_{t\in[T]} \left(Y_t(1)Y_{t+q}(1) + Y_t(0)Y_{t+q}(0)\right) \\ &+ \sum_{q=0}^{T/2} \beta_q^- \sum_{t\in[T]} \left(Y_t(1)Y_{t+q}(0) + Y_t(0)Y_{t+q}(1)\right) \end{aligned} \tag{16}$$

where $\beta_0^+ = 1/2 + O(1/T)$, $\beta_1^+ = -7/8 + O(1/T)$, and $\beta_q^+ = \frac{q^2-3q-12}{2^{q+3}} + O(1/T)$; $\beta_0^- = 0$, $\beta_1^- = -7/8 + O(1/T)$ and $\beta_q^- = \frac{q-4}{2^{q+2}} + O(1/T)$.

We leave the proof in Appendix D. For every $q \geq 1$, the coefficients still carry an $O(1/T)$ term. The reason is two-fold: first, there are T cross-moment pairs $Y_t(1)Y_{t+q}(1)$, and their combined contribution cannot be ignored; second, the constant factor (e.g., $\frac{q^2-3q-12}{2^{q+3}}$) diminishes more quickly as q approaches $T/2$. Nonetheless, when q is small, the constant component becomes our primary consideration. From the signs of the coefficients $\boldsymbol{\beta}^+$ and $\boldsymbol{\beta}^-$, we can see that $\widehat{\tau}^*$ involves the following tradeoff: on the one hand, it increases total variance by placing greater weight on each period's own second moments (i.e., $Y_t^2(1)$, $Y_t^2(0)$); on the other hand, it decreases total variance by reducing the influence of the cross moments between local periods (e.g., $Y_t(1)Y_{t+1}(1)$).

We introduce some additional notations to further reformulate this expression. We define the finite sample lag-q correlation of outcomes:

$$\rho_q(w) = \frac{\sum_{t\in[T]} \left(Y_t(w) - \bar{Y}(w)\right)\left(Y_{t+q}(w) - \bar{Y}(w)\right)}{\sum_{t\in[T]} \left(Y_t(w) - \bar{Y}(w)\right)^2}. \tag{17}$$

Similarly, we also define the finite sample lag-q correlation of treatment effect:

$$\rho_q^\tau = \frac{\sum_{t\in[T]} \left(\tau_t - \tau\right)\left(\tau_{t+q} - \tau\right)}{\sum_{t\in[T]} \left(\tau_t - \tau\right)^2}. \tag{18}$$

COROLLARY 1. *The difference of variance* (16) *can be rewritten as*

$$\begin{aligned} &T\left(\mathrm{Var}(\widehat{\tau}^*) - \mathrm{Var}(\widehat{\tau})\right) \\ &= \mathrm{Var}(Y_t(1)) \left(\sum_{q=0}^{T/2} (\beta_q^+ + \beta_q^-)\rho_q(1)\right) + \mathrm{Var}(Y_t(0)) \left(\sum_{q=0}^{T/2} (\beta_q^+ + \beta_q^-)\rho_q(0)\right) - \mathrm{Var}(\tau_t) \left(\sum_{q=0}^{T/2} \beta_q^- \rho_q^\tau\right). \end{aligned} \tag{19}$$

[2] We let $t+q \to t+q-T$ if $t+q > T$.

We leave the proof in Appendix E. This expression includes variance of treatment outcomes, variance of control outcomes and variance of treatment effect. In spirit, this echoes the classical representation in the analysis of experiments (Neyman 1923, Imbens and Rubin 2015). It is normal to believe that the lag-q correlation $\rho_q(w)$ diminishes in q, and it is about the rate that makes a difference. Unfortunately, as we do not impose any parametric assumptions on the potential outcomes, it is challenging to know the lag-q correlations $\rho_q(w)$ and ρ_q^τ for all q.

We provide a practical strategy (i) exploits local correlations and (ii) disregards correlations of treatment effect. Focusing on local correlations is justified because the magnitudes of the coefficients β_q^+ and β_q^- so performance is driven almost entirely by correlations among outcomes in nearby periods. We ignore treatment effect correlations partly because they cannot be estimated, and partly because, in most applications, treatment effect variability is negligible compared to outcome variability. For example, we can consider $q \leq 1$ when we essentially check

$$\mathrm{Var}(Y_t(1)) \left((\beta_0^+ + \beta_0^-) + (\beta_1^+ + \beta_1^-)\rho_1(1)\right) + \mathrm{Var}(Y_t(0)) \left((\beta_0^+ + \beta_0^-) + (\beta_1^+ + \beta_1^-)\rho_1(0)\right), \tag{20}$$

which is equivalent to

$$\mathrm{Var}(Y_t(1)) \left(\frac{1}{2} - \frac{7}{4}\rho_1(1)\right) + \mathrm{Var}(Y_t(0)) \left(\frac{1}{2} - \frac{7}{4}\rho_1(0)\right). \tag{21}$$

To this end, the first method is to estimate the variances and the correlations using the experiment data, and plug in the expression to check the sign. The second method is to infer the correlations from the historical data, by assuming that the treatment assignment does not affect (or weakly affects) the correlations. In this way, we can see that it suffices to check $\frac{1}{2} - \frac{7}{4}\rho_1(w)$. Once the lag-1 correlation $\rho_1(w) \geq \frac{2}{7}$, the robust optimization under the uncertainty set $\mathcal{Y}_1^{\mathrm{obs}}$ is preferred; otherwise, the one under $\mathcal{Y}_0^{\mathrm{obs}}$ is preferred. If we further consider $q \leq 2$, then it suffices to check $\frac{1}{2} - \frac{7}{4}\rho_1(w) - \frac{9}{16}\rho_2(w)$, which further relaxes the requirement of the lag-1 correlation needed.

Remark. The variance analysis also points to alternative robust optimization approaches that can achieve even lower variance than either $\mathrm{Var}(\widehat{\tau}^*)$ or $\mathrm{Var}(\widehat{\tau})$. The idea is to reframe the objective to a weighted simple regret where we down-weight imputed outcomes and up-weight observed ones. This leads to the optimal decision rule constructed from a general weighted average of $\widehat{\tau}^*$ and $\widehat{\tau}$, with the best weights determined by the autocorrelation structure. We do not further explore this because (i) from the statistical inference perspective, it adds no new statistical innovation, and its analysis follows mostly the same algebra; (ii) from the robust optimization perspective, the goal of this section is to showcase that it is essentially a general approach including a classical solution and a non-classical solution, emphasizing how they relate and differ.

3.4. Risk Preference

The previous discussion focuses on the risk-neutral setting where one is indifferent between the worst-case reward and the worst-case loss. We highlight that (3) is not the only formulation of robust optimization. In principle, one can easily adjust the formulation based on the risk preference between reward and loss. Taking the refined uncertainty as an instance, with the worst-case loss $\mathcal{R}(\mathbb{Y}^*(1),1)$ and the worst-case reward $\mathcal{R}(\mathbb{Y}^*(0),0)$, we consider

$$\min\{\mathcal{R}(\mathbb{Y}^*(1),1)+\gamma,\mathcal{R}(\mathbb{Y}^*(0),0)\} \tag{22}$$

where $\gamma\in\mathbb{R}$. This is equivalent to the decision rule

$$Z^*=\begin{cases}1, & \widehat{\tau}^*>\gamma\\ 0, & \widehat{\tau}^*\leq\gamma\end{cases} \tag{23}$$

Typically, a positive γ is specified for risk-averse decision makers. That is, the new treatment is adopted only if the gap between the worst-case reward and the worst-case loss is more than γ. The choice of γ is similar to the choice of significance level that balances the Type I and II errors, in classical hypothesis testing, but is directly linked to the cost perspective. We will investigate the impact of this parameter in the numerical study later.

4. Numerical Study

In this section, we conduct a numerical study to demonstrate the performance of robust optimization approaches. Since we are in the causal context, we specify the following outcome model that governs the observed data and the counterfactual data:

$$Y_t(W_t)=\mu+\sin\left(\frac{\pi t}{6}\right)+W_t\tau_t+\epsilon_t. \tag{24}$$

We let the intercept $\mu=2$ and the residual $\epsilon_t\sim\mathrm{U}(-0.5,0.5)$. Later we will consider different choices of the treatment effect $\tau_t\in[-0.2,0.2]$. As a result, we have the support $Y_t(W_t)\in[0,4]$.

4.1. Errors and Regrets

Although our new approach does not theoretically control Type I error, we can still investigate its empirical performance in Type I and II errors, compared with the traditional hypothesis testing. We consider the one-sided null

$$H_0:\tau=\frac{1}{T}\sum_{t=1}^{T}(Y_t(1)-Y_t(0))\leq 0, \tag{25}$$

and we set the significance level to be 0.1. In this subsection, we fix $T=168$ (i.e. one week horizon if each period is one hour) and vary τ_t in (24) from -0.2 to 0.2.

We first estimate the treatment effect using (9) and test the null (25) with a p-value given by the finite population central limit theorem (Li and Ding 2017). The Type I error rates under the

non-positive treatment effect ($\tau \leq 0$) are shown in Figure 1, and these remain well controlled at the 0.1 level, while the Type II error rates for positive treatment effects ($\tau > 0$) are displayed in Figure 2. We then apply the robust optimization (22) with different uncertainty sets. The parameter γ, which reflects the risk preference, does not directly correspond to Type I error control. To ensure a fair comparison, we calibrate γ so that the Type I error rate under $\tau = 0$ hits 0.1, matching the hypothesis testing benchmark. In other words, we anchor the robust optimization approach towards the hypothesis testing at $\tau = 0$ and study the performance for $\tau < 0$ and $\tau > 0$. As suggested by the structural property in Theorem 1, for robust optimization with the naive uncertainty set, both Type I and II error rates coincide with those of the hypothesis testing (See Robust-C and P-Value in Figure 1 - 2). For robust optimization with the refined uncertainty set, both Type I and II error rates improve substantially (See Robust-L in Figure 1 - 2).

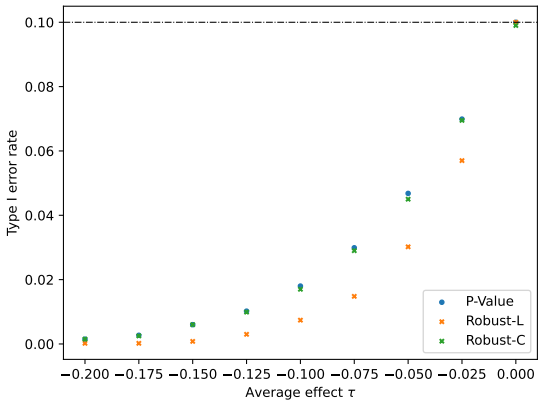


Figure 1 **Type I error under non-positive treatment effect**

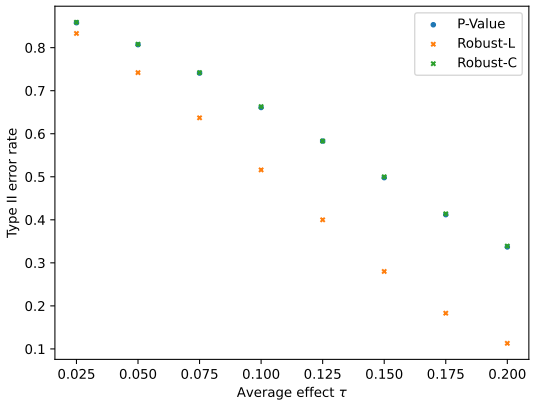


Figure 2 **Type II error under positive treatment effect**

Beyond error rates, we compute the actual regret based on the outcome model (24) by comparing decisions made under robust optimization and hypothesis testing. Figure 3 presents the average regret under the same setup in Figure 1 - 2, showing that the conservative nature of strict Type I error control results in low regret under negative effects, while the main differences appear under positive effects. Notably, the maximum regret of the hypothesis testing (0.074) is reduced by nearly 30% compared to the refined robust optimization approach (0.052). One may argue that this large gap is due to the fact that hypothesis testing prioritizes controlling false positives, often at the cost of increased conservativeness and lack of power. To further highlight the benefits, we consider a risk-neutral setting in Figure 4, where a significance level of 0.5 and $\gamma = 0$ are adopted, yielding a symmetric plot and a similar 31% reduction in maximum regret (from 0.0196 to 0.0134).

Based on our analysis in Section 3.3, we attribute the observed reduction in regret primarily to the strong temporal correlation in the outcome model (24) across consecutive periods. To further

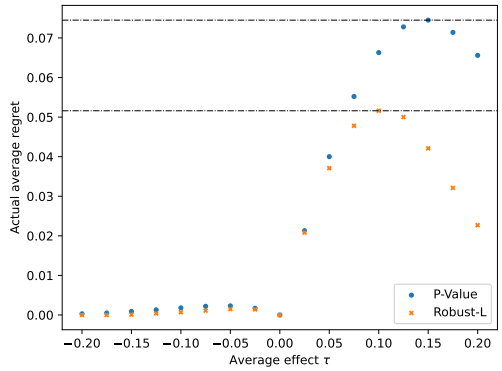


Figure 3 **Average regret at significance level 0.1**

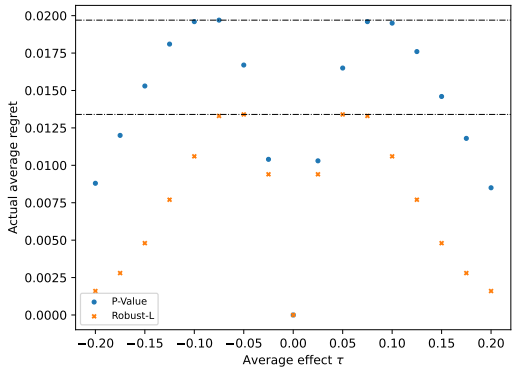


Figure 4 **Average regret at significance level 0.5**

investigate this effect, we conduct an additional simulation in which we vary the variance of the residuals $\epsilon_t \sim \mathrm{U}(-b, b)$ by changing the value of b. As b increases, the time series becomes noisier, making observations across periods more closely resemble i.i.d. samples. In such settings, robust optimization with a refined uncertainty set performs worse. Specifically, following the setup in Figure 4, we evaluate the expected regret with an average treatment effect fixed at $\tau = 0.1$. In Figure 5, we increase b from 0.5 to 4.5 and observe that Robust-L outperforms Robust-C when b is small. However, as b increases beyond 2, Robust-C begins to outperform Robust-L, though the performance gap remains relatively small even for large values of b. We then implement the practical strategy proposed in Section 3.3, which involves estimating the variances and covariances in (21) and checking their signs to determine which uncertainty set to use. This self-adaptive approach (Robust-C/L) effectively captures the strengths of both methods—matching the performance of Robust-C for large b and approaching that of Robust-L for small b. The minor performance loss observed is due to the conservative nature of our strategy: we rely solely on the lag-1 autocorrelation, and if it is insufficiently large, we default to Robust-C, even when higher-order autocorrelations (lag-p) might still favor Robust-L.

These simulation results indicate that robust optimization with a refined uncertainty set can outperform traditional hypothesis testing by offering better protection under negative effects while also enhancing exploitation under positive effects. Although the risk preference parameter γ captures the consideration of expected reward and loss, its mapping to Type I error control remains implicit. Thus, while we do not claim that our approach dominates hypothesis testing, the findings suggest that incorporating structure into the uncertainty set of the robust optimization can lead to substantial in terms of experimental power and decision regret.

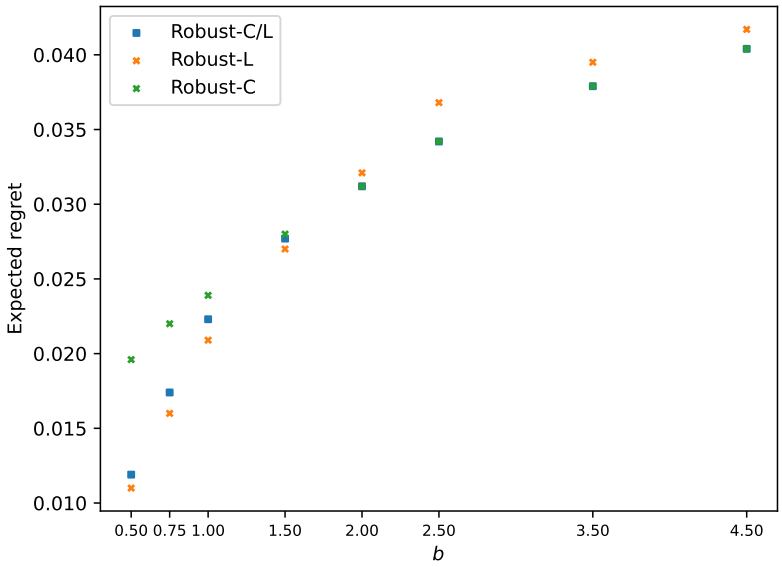


Figure 5 **Expected regret as the variance of independent residuals ϵ_t increases under $\tau = 0.1$ and $\gamma = 0$**

4.2. Two-stage Experimentation

Although we have shown the reduction of empirical Type I and II errors, the primary concern of our new approach is the lack of inherent Type I error control—a cornerstone of traditional hypothesis testing to provide statistical rigor. While the optimization framework excels at identifying promising treatments by directly solving a decision problem, its relaxed statistical guarantees necessitate further validation. Therefore, we propose a two-stage experimentation by dividing the experimental horizon into a screening phase (with optimization) and a confirmatory phase (with traditional statistical inference). The study seeks to illustrate how this hybrid design can enhance power in detecting true treatment effects while still controlling the overall Type I error rate. This two-stage structure is particularly appealing in resource-tight environments where allocating limited experimental resources among many experiments may dilute the power of tests.

Let there be K different experiments that one aims to test with a budget of T time periods.

1. In the first stage, all K experiments receive a uniform allocation of T_1 time periods. The risk-neutral robust optimization (12) is applied to each experiment based on the observed data. This step serves as an effective filter to select K^* promising candidates.
2. In the second stage, K^* candidates deemed promising in the first stage are then tested using conventional statistical methods (e.g. confidence intervals from finite-sample central limit theorem) with another uniform allocation of the remaining $T_2 = T - T_1$ time periods. This stage employs standard hypothesis testing procedures to make final decisions, ensuring that the overall Type I error rate is maintained.

The performance of the two-stage design is directly compared to that of the single-stage design, in which T periods are equally divided among all K experiments, followed by hypothesis testing. To ensure that both designs achieve the same overall Type I error control, we set the significance level for the two-stage design at 0.2. This adjustment accounts for the screening phase in the two-stage design, which has a 50% chance of selecting an experiment with $\tau = 0$ for the confirmatory phase; when combined, this controls the overall Type I error rate at 0.1. A formal statement and proof are provided in Appendix F.

In this simulation, we set $K = 10$ and $T = 336 \cdot K = 3360$ periods. We keep using the outcome model (24) for our data generation process. For each problem instance, we randomly generate K experiments with average treatment effects $\tau \sim N(0, 0.1)$ independently. Given these K experiments, we apply a specific design (either single-stage or two-stage) to allocate resources, generate realized assignments, perform inference, and verify correctness against the true effect τ of each experiment. By repeating this procedure over 5000 realizations, we summarize the error rates for all $K = 10$ experiments. Finally, we generate 100 problem instances and collect 1000 data points, each containing the treatment effect and the corresponding error rate. Figure 6 presents scatter plots of these 1000 data points for both the single-stage and two-stage designs. In the two-stage design, we consider two allocation schemes: one with $T_1 = 0.5T$ and another with $T_1 = 0.25T$. Our findings indicate that Type I errors are well controlled in all cases, while the Type II error is lower for the two-stage design compared to the single-stage design. Notably, the performance gap is most pronounced under moderate treatment effects (between 0.1 and 0.2), and the allocation with $T_1 = 0.25T$ slightly outperforms the one with $T_1 = 0.5T$.

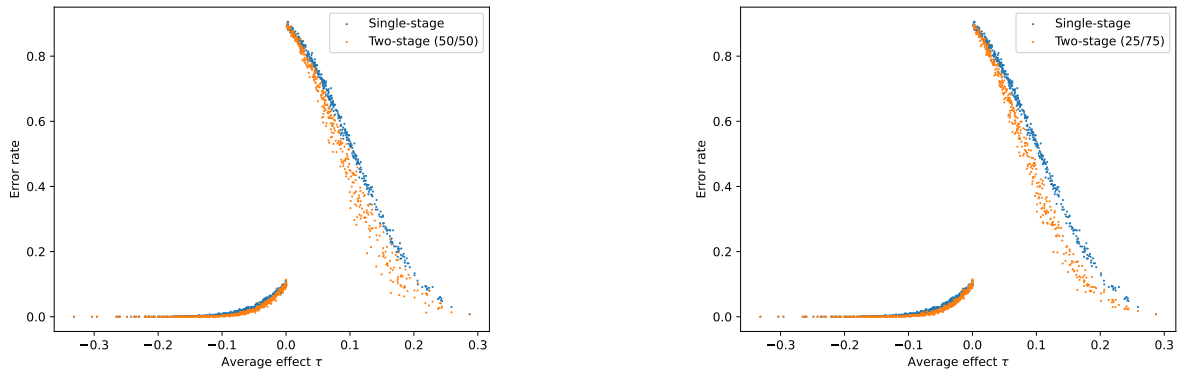


Figure 6 Comparison of error rates: single-stage and two-stage designs with $T_1 = 0.5T$ (left) and $T_1 = 0.25T$ (right)

Through this structured comparison, we provide empirical evidence supporting the benefit of our robust optimization approach in the decision analytics of experiments. By combining the exploratory

advantages of robust optimization with the inferential reliability of traditional statistics, we enhance detection power without compromising on error control—a compelling proposition in settings with multiple competing experiments and finite resources.

5. Extension: Carryover Effects

Switchback experiments can effectively reduce network interference among users, yet they often suffer from carryover effects (or temporal interference), where past treatments influence current outcomes (Cox 1958). As a result, Assumption (1) no longer holds (i.e., $Y_t(\mathbf{1}) \neq Y_t(1)$), which prompts us to revert to our general causal estimand:

$$\tau = \frac{1}{T}\sum_{t=1}^{T}\left(Y_t(\mathbf{1}) - Y_t(\mathbf{0})\right). \tag{26}$$

In this section, we extend our robust optimization approach to explicitly accommodate the presence of carryover effects. Prior literature has proposed two primary frameworks for integrating these effects into the design and analysis of switchback experiments (Bojinov and Shephard 2019, Bojinov et al. 2022, Hu and Wager 2022, Ni et al. 2023, Xiong et al. 2024, Jia et al. 2024): (i) restricting the duration of carryover effects to minimize variance, and (ii) characterizing the decay of carryover effects to study the bias-variance tradeoff for minimizing the mean squared error. Here, we leverage these insights and incorporate them into our robust optimization methodology.

5.1. Robust Optimization Formulation

We consider the design of experiments (Bojinov et al. 2022, Ni et al. 2023, Hu and Wager 2022, Jia et al. 2024) parameterized by the switching window size l. That is, for every l periods, we flip a fair coin to determine the treatment assignments during these l periods. Despite different estimation strategies in the literature, one common (and inevitable) element is that for each time period t, we only check the treatment assignments within its "neighborhood" instead of the entire history — for example, past b periods together with the current period. Once all the recent $b+1$ periods are assigned to the treatment group, we take the observed outcome as a good approximation for the outcome under global treatments (i.e., $Y_t(\mathbf{1}_{b+1}) \approx Y_t(\mathbf{1})$). To conduct the inference, we consider a truncated version of IPW estimator to estimate the global average treatment effect:

$$\widehat{\tau} = \frac{1}{T-b}\sum_{t=b+1}^{T} Y_t^{\text{obs}}\left(\frac{\mathbb{1}\{\boldsymbol{W}_{t-b:t} = \mathbf{1}_{b+1}\}}{\mathbb{P}(\boldsymbol{W}_{t-b:t} = \mathbf{1}_{b+1})} - \frac{\mathbb{1}\{\boldsymbol{W}_{t-b:t} = \mathbf{0}_{b+1}\}}{\mathbb{P}(\boldsymbol{W}_{t-b:t} = \mathbf{0}_{b+1})}\right). \tag{27}$$

For the robust optimization formulation, we also employ the parameter b to define the decision variables and the uncertainty set. Note that within a neighborhood of b periods, there are $2(b+1)$ possible assignment patterns can be generated by the design, denoted as

$$\mathcal{W}_b = \{(\mathbf{1}_m, \mathbf{0}_{b+1-m}), \forall m \in [b+1]\}. \tag{28}$$

Then we define $2(b+1)$ decision variables for each period t: $\{y_t(\boldsymbol{w}), \forall \boldsymbol{w} \in \mathcal{W}_b\}$. The objective function (i.e. regret) is adjusted as

$$\mathcal{R}(\mathbb{Y}, Z) = \frac{1}{T-b} \sum_{t=b+1}^{T} y_t(\mathbf{1}_{b+1} - \boldsymbol{Z}_{b+1}) - y_t(\boldsymbol{Z}_{b+1}). \tag{29}$$

For the feasible region, it is straightforward to inherit the refined uncertainty set $\mathcal{Y}_1^{\text{obs}}$. Specifically, for any decision variable $y_t(\boldsymbol{w})$,

$$y_t(\boldsymbol{w}) = Y_t^{\text{obs}}, \boldsymbol{W}_{t-b:t}^{\text{obs}} = \boldsymbol{w}; |y_{t+1}(\boldsymbol{w}) - y_t(\boldsymbol{w})| \leq L, \boldsymbol{W}_{t-b:t}^{\text{obs}} \neq \boldsymbol{w}. \tag{30}$$

The main difference is that without carryover effects, continuity is applied solely to the treatment group $(y_t(1), y_{t+1}(1))$ and the control group $(y_t(0), y_{t+1}(0))$. In contrast, when carryover effects are present, continuity extends to any fixed combination of assignments of length $b+1$. For example, if we set $b=3$ and consider the pattern $\boldsymbol{w}=(1,1,0,0)$, it implies that for two consecutive periods t and $t+1$, the potential outcomes under the same assignment pattern $\boldsymbol{w}=(1,1,0,0)$ do not differ significantly, though they cannot be observed simultaneously.

We have now generalized the robust formulation from the no-carryover scenario to the carryover setting. However, we note that the objective function only involves the potential outcomes regarding assignment patterns $\boldsymbol{w} \in \{\mathbf{1}_{b+1}, \mathbf{0}_{b+1}\}$. As a result, for $\boldsymbol{W}_{t-b:t}^{\text{obs}} \notin \{\mathbf{1}_{b+1}, \mathbf{0}_{b+1}\}$, the corresponding observation $Y_t(\boldsymbol{W}_{t-b:t}^{\text{obs}})$ does not affect the objective, rendering it uninformative for decision making. To harness the potential value of more observations for better data efficiency, we aim to introduce additional constraints to the uncertainty set that capture the underlying structure of carryover effects as outlined in the following assumption:

ASSUMPTION 4 **(Diminishing carryover effects)**. *At each period t, with $1 \leq m \leq b$, the potential outcomes satisfy the following inequalities for any $\boldsymbol{w}$:*

$$\begin{aligned} &|Y_t(\boldsymbol{w}, \mathbf{1}_m, \mathbf{0}_{b+1-m}) - Y_t(\boldsymbol{w}, \mathbf{1}_{m-1}, \mathbf{0}_{b+2-m})| \geq |Y_{t+1}(\boldsymbol{w}, \mathbf{1}_m, \mathbf{0}_{b+2-m}) - Y_{t+1}(\boldsymbol{w}, \mathbf{1}_{m-1}, \mathbf{0}_{b+3-m})|, \\ &|Y_t(\boldsymbol{w}, \mathbf{0}_m, \mathbf{1}_{b+1-m}) - Y_t(\boldsymbol{w}, \mathbf{0}_{m-1}, \mathbf{1}_{b+2-m})| \geq |Y_{t+1}(\boldsymbol{w}, \mathbf{0}_m, \mathbf{1}_{b+2-m}) - Y_{t+1}(\boldsymbol{w}, \mathbf{0}_{m-1}, \mathbf{1}_{b+3-m})|. \end{aligned} \tag{31}$$

This assumption requires that the carryover effect of past treatment on future outcomes diminishes. For example, if we let $b=3, m=2$, then $Y_t(\boldsymbol{w}, 1, 1, 0, 0) - Y_t(\boldsymbol{w}, 1, 0, 0, 0)$ captures the carryover effect on period t from a latest treatment at $t-2$, whose size should be no smaller than $Y_{t+1}(\boldsymbol{w}, 1, 1, 0, 0, 0) - Y_{t+1}(\boldsymbol{w}, 1, 0, 0, 0, 0)$, its carryover effect on period $t+1$. This is similar to the rapid mixing condition proposed in the literature (Hu and Wager 2022, Jia et al. 2024), but without imposing that the system stabilizes exponentially.

To integrate the assumption into the uncertainty set, as the decision variables are only defined within a neighborhood of b, we construct the new uncertainty set $\mathcal{Y}_2^{\text{obs}}$ by adding to $\mathcal{Y}_1^{\text{obs}}$ the monotone constraints that for each period $b+1 \le t \le T-1$ and $1 \le m \le b$:

$$\begin{aligned} |y_t(\mathbf{1}_m, \mathbf{0}_{b+1-m}) - y_t(\mathbf{1}_{m-1}, \mathbf{0}_{b+2-m})| &\ge |y_{t+1}(\mathbf{1}_m, \mathbf{0}_{b+2-m}) - y_{t+1}(\mathbf{1}_{m-1}, \mathbf{0}_{b+3-m})|, \\ |y_t(\mathbf{0}_m, \mathbf{1}_{b+1-m}) - y_t(\mathbf{0}_{m-1}, \mathbf{1}_{b+2-m})| &\ge |y_{t+1}(\mathbf{0}_m, \mathbf{1}_{b+2-m}) - y_{t+1}(\mathbf{0}_{m-1}, \mathbf{1}_{b+3-m})|. \end{aligned} \tag{32}$$

Then it remains to solve $\max_{\mathbb{Y} \in \mathcal{Y}_2^{\text{obs}}} \mathcal{R}(\mathbb{Y}, Z)$ for the decision $Z \in \{0,1\}$. Unfortunately, this is no longer a linear program because the new inequalities introduced are nonlinear. In light of this difficulty, we introduce auxiliary binary variables and reformulate the robust optimization into a mixed-integer linear program:

$$\begin{aligned} \max \quad & \frac{1}{T-b} \sum_{t=b+1}^{T} \left(y_t(\mathbf{0}_{b+1}) - y_t(\mathbf{1}_{b+1})\right) \\ \text{s.t.} \quad & y_t(\boldsymbol{W}_{t-b:t}^{\text{obs}}) = Y_t^{\text{obs}}, \quad \forall\ b+1 \le t \le T \\ & -L \le y_t(\boldsymbol{w}) - y_{t+1}(\boldsymbol{w}) \le L, \quad \forall\ b+1 \le t \le T-1, \boldsymbol{w} \in \mathcal{W}_b, \boldsymbol{W}_{t-b:t}^{\text{obs}} \ne \boldsymbol{W}_{t-b+1:t+1}^{\text{obs}} \\ & -(1-s_t(\boldsymbol{w}))B \le y_t(\boldsymbol{w}) - y_t(\boldsymbol{w}) \le s_t(\boldsymbol{w})B, \quad \forall\ b+1 \le t \le T-1, \boldsymbol{w} \in \mathcal{W}_b \\ & -(1-s_t(\boldsymbol{w}))B \le (y_t(\boldsymbol{w}) - y_t(\boldsymbol{w})) - (y_{t+1}(\boldsymbol{w}) - y_{t+1}(\boldsymbol{w})) \le s_t(\boldsymbol{w})B, \ \forall\ b+1 \le t \le T-1, \boldsymbol{w} \in \mathcal{W}_b \\ & -(1-s_t(\boldsymbol{w}))B \le (y_t(\boldsymbol{w}) - y_t(\boldsymbol{w})) - (y_{t+1}(\boldsymbol{w}) - y_{t+1}(\boldsymbol{w})) \le s_t(\boldsymbol{w})B, \ \forall\ b+1 \le t \le T-1, \boldsymbol{w} \in \mathcal{W}_b \end{aligned}$$

5.2. Numerical Study

We specify the following general form of time series models (Scott and Varian 2014, Brodersen et al. 2015) as the data generation process:

$$\begin{aligned} Y_t(\boldsymbol{w}_{1:T}) &= \alpha_t + s_t + \lambda_t^Y w_t + \varepsilon_t, \\ \alpha_t &= \mu_t + \rho_t \alpha_{t-1} + \lambda_t^\alpha w_t + \eta_t, \end{aligned} \tag{33}$$

where α_t is the unobserved hidden state of the dynamic system, s_t incorporates the seasonality, and $\lambda_t^Y w_t$ captures the instantaneous treatment effect on the outcome. The unobserved hidden state α_t evolves over time, depending on the drift μ_t, the previous state $\rho_t \alpha_{t-1}$ (auto-regressive) and the intervention $\lambda_t^\alpha w_t$. From this characterization, one can see that the effect λ_t^α is carried over to future periods through the hidden state. In the extreme case, the first period treatment w_1 may impact the outcome at the last period $Y_T(\boldsymbol{w}_{1:T})$, complicating the identification of the global treatment effect.

For our specification, we set the drift $\mu_t = 0$ and define the seasonal component as $s_t = \sin\left(\frac{\pi t}{12}\right) + 2.5$. The residuals η_t, ϵ_t are assumed to be independently and uniformly sampled from $U(-0.5, 0.5)$. For the treatment effect, we fix the direct effect from the current period $\lambda_t^Y = 0.4$ and vary the indirect effect λ_t^α. We are particularly interested in the case when λ_t^α is negative, since having an indirect

effect with an opposite sign to λ_t^Y can critically influence the decision-making. Finally, by considering a stationary auto-regressive parameter $\rho_t = 0.5$, the global average treatment effect can be explicitly derived as $\tau = 0.4 + \lambda_t^\alpha / 0.5$, which allows us to easily identify both type I and type II errors.

5.2.1. Errors and Regrets Similar to Section 4.1, we first examine how robust optimization performs in reducing errors and regrets. We fix $T = 336$ while maintaining a significance level of 0.1. To represent different degrees of carryover effects, we vary λ_t^α from -0.4 to 0. According to our outcome model, the global treatment effect $\tau \leq 0$ under $\lambda_t^\alpha \leq -0.2$ so we evaluate the Type I error and $\tau > 0$ under $\lambda_t^\alpha > -0.2$ so we evaluate the Type II error.

Two notable differences arise compared to the case with no carryover effects. First, under hypothesis testing with $\tau \leq 0$, the Type I error is only approximately controlled because the bias may never fully vanish. Nonetheless, we still calibrate γ so that the Type I error from the robust optimization approach aligns with the empirical error of the hypothesis testing method. Second, the design parameters b and l not only affect the bias-variance trade-off under the truncated IPW estimator (27) but also influence the performance of the robust optimization approach by altering the decision variables and structural constraints. Consequently, we consider several setups for b and l to facilitate a better understanding of the impact of experimental design.

In Table 1, the column for $\tau = 0$ displays the Type I errors under various design parameters, while the remaining columns report the Type II errors. We first observe that larger b gives closer Type I error control to 0.1, but generally larger Type II error for both the hypothesis testing approach with p-values and the robust optimization approach. The performance of robust optimization consistently dominates the hypothesis testing approach, and introducing the monotone constraints for diminishing carryover effects further improves it, but the effectiveness of two robust optimization approaches depends on the design parameters.

- As the switching window l increases, the performance gap between P-Value and Robust-L becomes less pronounced. This occurs because the optimal solution under the continuity constraints can be interpreted by imputing missing treatment (or control) outcomes (i.e.. $\boldsymbol{W}_{t-b:t+1}^{\text{obs}} \notin \{\mathbf{1}_{b+1}, \mathbf{0}_{b+1}\}$) by extrapolating from the two nearest observed outcomes, as established in Theorem 2. With a larger switching window, the intervals of consecutive missing treatment outcomes lengthen, meaning that the imputed values rely on observations that are further apart. Consequently, the benefit of the continuity constraints diminishes, since extrapolation from distant periods tends to be less informative.
- As the neighborhood size b increases, the performance gap between Robust-L and Robust-L-M becomes more pronounced. A larger b results in a greater number of missing treatment (or control) outcomes (i.e., $\boldsymbol{W}_{t-b:t+1}^{\text{obs}} \notin \{\mathbf{1}_{b+1}, \mathbf{0}_{b+1}\}$) and these missing observations $Y_t(\boldsymbol{W}_{t-b:t+1}^{\text{obs}})$

are not incorporated in Robust-L. In contrast, Robust-L-M leverages monotone constraints to account for diminishing carryover effects, effectively linking observations over time, which becomes increasingly valuable with a larger neighborhood size.

It is straightforward to compare actual regrets by scaling the errors by the effect size; however, since the resulting insights are similar, we omit that analysis here. Instead, we highlight an interesting observation regarding the interplay between the experimental design and the decision framework. For example, when $b=3$, hypothesis testing generally produces smaller Type II errors under $l=6$ than under $l=4$. In contrast, under robust optimization, this relationship is reversed. This reversal implies how the optimal experimental design may depend on the decision framework in use, much like its dependency on the specific causal effect of interest.

		Method	$\tau=0$	$\tau=0.1$	$\tau=0.2$	$\tau=0.3$	$\tau=0.4$
$b=2$	$l=3$	P-Value	0.128	0.761	0.601	0.415	0.256
		Robust-L	0.128	0.680	0.426	0.200	0.071
		Robust-L-M	0.128	**0.679**	**0.422**	**0.193**	**0.063**
	$l=4$	P-Value	0.121	0.761	0.606	0.426	0.266
		Robust-L	0.121	0.740	0.547	0.317	0.156
		Robust-L-M	0.121	**0.734**	**0.526**	**0.298**	**0.139**
$b=3$	$l=4$	P-Value	0.113	0.803	0.667	0.497	0.337
		Robust-L	0.113	0.764	0.576	0.377	0.207
		Robust-L-M	0.113	**0.747**	**0.548**	**0.328**	**0.160**
	$l=6$	P-Value	0.111	0.782	0.642	0.472	0.304
		Robust-L	0.111	0.777	0.630	0.454	0.289
		Robust-L-M	0.111	**0.768**	**0.607**	**0.422**	**0.252**

Table 1 **Comparison of error rates under different treatment effects with different design parameters**

5.2.2. Two-stage Experimentation We extend the idea of two-stage experimentation to the case with carryover effects by choosing the setup of $b=2, l=4$. We follow the same procedure in Section 4.2, but change the data generation process to the time series model (33). Since we have $\tau=0$ under $\lambda_t^\alpha=-0.2$, when generating K experiments, we randomly sample $\lambda_t^\alpha \sim N(-0.2, 0.1)$, which is equivalent to $\tau \sim N(0, 0.2)$. Similar to Figure 6, we present the scatter plots of error rates for both the single-stage and two-stage designs in Figure 7 under $T_1=0.5T$ and $T_1=0.25T$. Because of estimation bias, both approaches yield Type I error rates above the nominal 0.1 level; the excess is slightly larger for the two-stage design, since the robust optimization more readily flags small, positively biased effects as significant. Even so, the two-stage design markedly increases power. For treatments of moderate size (effect 0.2–0.3), it reduces Type II errors by nearly 50%.

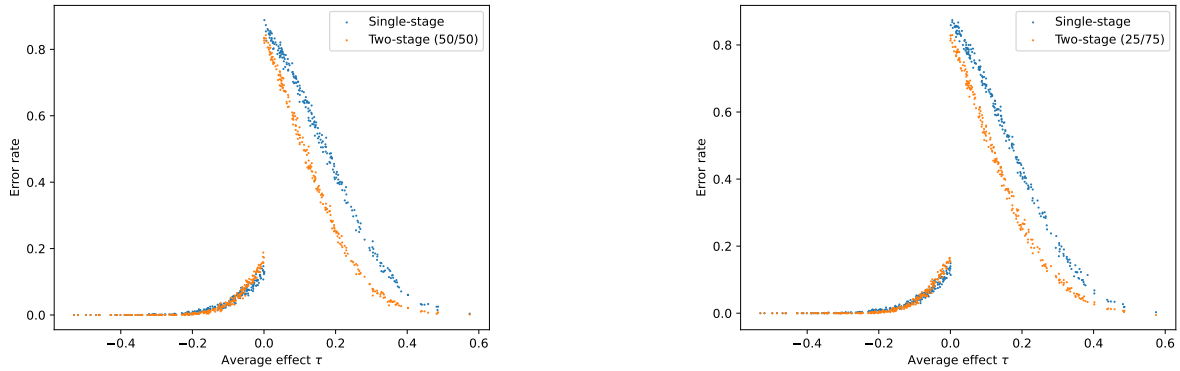


Figure 7 **Comparison of error rates in the presence of carryover effects: single-stage and two-stage designs with $T_1 = 0.5T$ (left) and $T_1 = 0.25T$ (right)**

6. Concluding Remarks and Discussions

In this paper, we reframe decision making for switchback experiments as a minimax choice between treatment and control rather than a classical hypothesis test. By modeling the unobserved counterfactual outcomes as adversarial, we derive two closed-form rules—one that relies only on naive averages and another refined one that also exploits Lipschitz-smoothness across time. In simulation, the refined one reduces expected regret and Type II error while preserving Type I control. We extend the framework to handling carryover effects and incorporate additional constraints that build on the assumption of diminishing carryover effects. Overall, robust optimization serves as a practical, low-overhead complement to traditional inference, acting as an effective first-stage filter that protects business metrics and conserves scarce experimental resources. That being said, we discuss several limitations and considerations, explaining why they are not addressed in this paper and how future works can be developed.

6.1. The Design of Experiments

Our analysis takes the experimental design as given and focuses on decision-making with the resulting data. The rich interplay between design and analysis has been widely discussed in the literature (Dean and Voss 1999, Eckles et al. 2017, Athey and Imbens 2017, Montgomery 2017, Ni 2025). Robust optimization, however, has design implications of its own. Under a classical completely randomized design, the two robust formulations we study behave differently. Now consider a mirror (matched-pair) design: flip a fair coin in period 1 to assign treatment W_1, and in period 2 force the opposite assignment $W_2 = 1 - W_1$; repeat this pattern every two periods. By pairing local observations, this design narrows the gap between the two robust rules—even the classical IPW estimator gains efficiency when consecutive outcomes are similar. Yet when carryover effects are present, the influence of alternative designs on general robust formulations becomes less transparent. A thorough design-oriented investigation remains an open avenue for future work.

6.2. The Function of Robust Optimization

Our premise of this paper is that robust optimization offers a flexible, powerful alternative for decision analytics in switchback experiments by incorporating the problem structure, without any estimation or prediction. In fact, for the naive uncertainty set, a uniform bound $[0, C]$ may not be appropriate due to the non-stationarity such as peak and non-peak hours. We can extend it by introducing an outcome prediction (e.g., fitting a time series model) and assume that the true potential outcome is bounded around the predicted outcome.

Moreover, the formulation is conceptually simple compared to recent advances in distributionally robust optimization and robust satisficing (Delage and Ye 2010, Kuhn et al. 2019, Rahimian and Mehrotra 2022, Long et al. 2023, Wang et al. 2023). Those methods typically assume an underlying true distribution and make robust decisions based on an empirical distribution. We depart from that paradigm, adopting a design-based perspective in which potential outcomes are fixed but unknown—an outlook particularly apt for switchback experiments with complex temporal dependence and carryover. Exploring how modern distributionally robust techniques can be adapted to this setting is an intriguing direction for future research.

6.3. The Impact of Contextual Information

Our robust decision rule Z is designed to determine whether a treatment should be deployed across all periods, rather than characterizing a robust individualized treatment rule based on the covariates X_t observed in each period. Technically, we could redefine the potential outcomes to be functions of X_t, and extend the robust optimization framework to define time-dependent decisions Z_t that depend on X_t (Stoye 2012). However, we set aside this direction for two main reasons. First, in many switchback experiments, the treatment and control arms correspond to fully specified policies—such as two pricing strategies or assignment mechanisms with different parameters. Each policy inherently accounts for covariates and generates outputs accordingly. Thus, analyzing such experiments is akin to off-policy evaluation (Farias et al. 2022), where the goal is to compare entire policies rather than optimize an individualized intervention at each time step. This setting differs fundamentally from classical interventions, such as deciding whether to administer a drug to a patient or offer a coupon to a user, which typically involve one-off, context-dependent actions. Second, even in traditional intervention settings, defining an individualized treatment rule over time is highly complex when there are carryover effects. A treatment at one time period can affect outcomes in future periods, making it unclear how to define the "optimal" decision at any single point in time—especially when future covariates are unobserved and future decisions are yet to be made. One potential alternative is to model the problem as a Markov Decision Process (MDP), where actions are chosen at each state (e.g., based on current covariates) to maximize the value function.

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Online Appendix

A. Proof of Theorem 1

We first derive the worst-case solution and the worst-case regret for the decision $Z=0$. It is straightforward that for the unobserved potential outcomes, the worst-case scenario should increase the outcome $y_t(1)$ under treatment as much as possible, and decrease the outcome $y_t(0)$ under control as much as possible. Under the bounded support constraint, this translates into $y_t(1)=C, \forall W_t^{\text{obs}} \neq 1$ and $y_t(0)=0, \forall W_t^{\text{obs}} \neq 0$. In this way, the worst-case reward becomes

$$\mathcal{R}(\mathbb{Y}^*(0),0)=\frac{1}{T}\sum_{t=1}^{T}(\mathbb{1}\{W_t=1\}\cdot Y_t(1)+\mathbb{1}\{W_t=0\}\cdot C)-(\mathbb{1}\{W_t=0\}\cdot Y_t(0)+\mathbb{1}\{W_t=1\}\cdot 0) \quad (34)$$

which is equivalent to $\frac{1}{2}(\widehat{\tau}+C)$. Similarly, for the decision $Z=1$, the worst-case loss becomes

$$\mathcal{R}(\mathbb{Y}^*(1),1)=\frac{1}{T}\sum_{t=1}^{T}(\mathbb{1}\{W_t=0\}\cdot Y_t(0)+\mathbb{1}\{W_t=1\}\cdot C)-(\mathbb{1}\{W_t=1\}\cdot Y_t(1)+\mathbb{1}\{W_t=0\}\cdot 0) \quad (35)$$

which is equivalent to $\frac{1}{2}(C-\widehat{\tau})$. Thus, the decision problem is equivalent to checking

$$\mathcal{R}(\mathbb{Y}^*(0),0)-\mathcal{R}(\mathbb{Y}^*(1),1)=\widehat{\tau}\geq 0. \quad (36)$$

B. Proof of Theorem 2

Similar to the proof in Appendix A, we first derive the worst-case solution and the worst-case regret for the decision $Z=0$.

To achieve the worst-case regret, we should increase the outcome $y_t(1)$ under treatment as much as possible given the Lipschitz condition. Let us consider two periods t and t' such that $W_t=1, \boldsymbol{W}_{t+1:t'-1}=\mathbf{0}, W_{t'}=1$. The worst-case solution can then be generated recursively. If $y_t^*(1)=Y_t^{\text{obs}}(1)\leq Y_{t'}^{\text{obs}}(1)=y_{t'}^*(1)$, then we have $y_{t+1}^*(1)=y_t^*(1)+L$; otherwise, we have $y_{t'-1}^*(1)=y_{t'}^*(1)+L$. According to different scenarios, we reset $t=t+1$ or $t'=t'-1$ and repeat the procedure until $t+1=t'$. During the process, the key is to know which side (i.e., smaller t or larger t') of the Lipschitz constraint should be binding, which is determined by the gap between $\Delta_{t,t'}(1)=y_{t'}^*(1)-y_t^*(1)$. When $t'-t$ is odd, the number of times when the smaller t is binding is given by

$$m_{t,t'}^{+}(1)=(t'-t-1)/2+\lfloor(\Delta_{t,t'}(1)/L+1)/2\rfloor; \quad (37)$$

when $t'-t$ is even, it is given by

$$m_{t,t'}^{+}(1)=(t'-t)/2+\lfloor(\Delta_{t,t'}(1)/L)/2\rfloor. \quad (38)$$

Then the worst-case outcomes under treatment from the original $t+1$ to $t'-1$ satisfy

$$\begin{aligned}\sum_{s=t+1}^{t'-1} y_s^*(1) =& Y_t^{\text{obs}}(1)\cdot m_{t,t'}^{+}(1)+(m_{t,t'}^{+}(1)+1)\cdot m_{t,t'}^{+}(1)\cdot L/2 \\ &+Y_{t'}^{\text{obs}}(1)\cdot(t'-t-1-m_{t,t'}^{+}(1))+((t'-t-m_{t,t'}^{+}(1)))\cdot(t'-t-1-m_{t,t'}^{+}(1))\cdot L/2\end{aligned} \quad (39)$$

We should also decrease the outcome $y_t(0)$ under control as much as possible given the Lipschitz condition. Let us consider two periods t and t' such that $W_t = 0, \boldsymbol{W}_{t+1:t'-1} = \mathbf{1}, W_{t'} = 0$. We can follow a similar argument above by changing $y^*_{t+1}(1) = y^*_t(1) + L$ to $y^*_{t+1}(0) = y^*_t(0) - L$, and

$$m^-_{t,t'}(1) = (t' - t - 1)/2 + \lceil (-\Delta_{t,t'}(1)/L - 1)/2 \rceil. \tag{40}$$

Then the worst-case outcomes under control satisfy

$$\begin{aligned} \sum_{s=t+1}^{t'-1} y^*_s(0) =& Y^{\text{obs}}_t(0) \cdot m^-_{t,t'}(1) - (m^-_{t,t'}(1) + 1) \cdot m^-_{t,t'}(1) \cdot L/2 \\ &+ Y^{\text{obs}}_{t'}(0) \cdot (t' - t - 1 - m^-_{t,t'}(1)) - ((t' - t - m^-_{t,t'}(1))) \cdot (t' - t - 1 - m^-_{t,t'}(1)) \cdot L/2. \end{aligned} \tag{41}$$

Finally, we obtain the worst-case regret

$$\mathcal{R}(\mathbb{Y}^*(0), 0) = \frac{1}{T} \sum_{t=1}^{T} (y^*_t(1) - y^*_t(0)) \tag{42}$$

Next we derive the worst-case solution and the worst-case regret $(\mathbb{Y}^*(1), 1)$ for the decision $Z = 1$. We still consider the worst-case outcomes under treatment first. They share the same outcomes for those observed periods, but differ in unobserved ones. Specifically, we should decrease the outcome $y_t(1)$ as much as possible given the Lipschitz condition. Thus, for the same interval between t and t' as above, we have

$$\begin{aligned} \sum_{s=t+1}^{t'-1} y^*_s(1) =& Y^{\text{obs}}_t(1) \cdot m^-_{t,t'}(1) - (m^-_{t,t'}(1) + 1) \cdot m^+_{t,t'}(1) \cdot L/2 \\ &+ Y^{\text{obs}}_{t'}(1) \cdot (t' - t - 1 - m^-_{t,t'}(1)) - ((t' - t - m^-_{t,t'}(1))) \cdot (t' - t - 1 - m^-_{t,t'}(1)) \cdot L/2. \end{aligned} \tag{43}$$

Therefore, during this interval with unobserved treatment outcomes, the sum of (39) and (43) contributes to the difference $\mathcal{R}(\mathbb{Y}^*(0), 0) - \mathcal{R}(\mathbb{Y}^*(1), 1)$. By checking the definitions, the key property we need is that $m^+_{t,t'}(1) + m^-_{t,t'}(1) = t' - t - 1$. Then the sum becomes

$$(Y^{\text{obs}}_t(1) + Y^{\text{obs}}_{t'}(1)) \cdot (t' - t - 1) \tag{44}$$

where the parts with L cancel out. This quantity is equivalent to

$$2 \sum_{s=t+1}^{t'-1} Y^\dagger_s(1), \quad Y^\dagger_s(1) = \frac{s - t}{t' - t} Y^{\text{obs}}_t(1) + \frac{t' - s}{t' - t} Y^{\text{obs}}_{t'}(1). \tag{45}$$

For the periods with observed outcomes, $2 \sum_{t=1}^{T} \mathbb{1}\{W_t = 1\} Y_t(1)$ contributes to the difference $\mathcal{R}(\mathbb{Y}^*(0), 0) - \mathcal{R}(\mathbb{Y}^*(1), 1)$.

The worst-case outcomes under control are symmetric, and finally, we can write down the difference between the worst-case reward and the worst-case control:

$$\begin{aligned}\mathcal{R}(\mathbb{Y}^*(0),0)-\mathcal{R}(\mathbb{Y}^*(1),1) = &\frac{2}{T}\sum_{t=1}^{T}(\mathbb{1}\{W_t=1\}Y_t(1)-\mathbb{1}\{W_t=0\}Y_t(0)) \\ &+\frac{2}{T}\sum_{t=1}^{T}(\mathbb{1}\{W_t=0\}Y_t^\dagger(1)-\mathbb{1}\{W_t=1\}Y_t^\dagger(0))\end{aligned} \tag{46}$$

Thus, given our definition of $\widehat{\tau}^*$, the decision problem is equivalent to checking

$$\mathcal{R}(\mathbb{Y}^*(0),0)-\mathcal{R}(\mathbb{Y}^*(1),1)=\widehat{\tau}^*\geq 0. \tag{47}$$

C. Proof of Proposition 1

We note that the unbiasedness of $\widehat{\tau}$ is a well-known result, so we omit the proof here and focus on the analysis of $\widehat{\tau}^*$. We first rewrite it into

$$\widehat{\tau}^*=\frac{1}{2}\widehat{\tau}+\frac{1}{T}\sum_{t\in[T]}\big(\mathbb{1}\{W_t=0\}Y_t^\dagger(1)-\mathbb{1}\{W_t=1\}Y_t^\dagger(0)\big) \tag{48}$$

For convenience, we let $\widehat{\tau}^\dagger$ denote the second part. Then it suffices to show $\mathbb{E}[\widehat{\tau}^\dagger]=\frac{1}{2}\tau$. We prove this by investigating how a certain outcome $Y_t(1)$ appears in $\widehat{\tau}^\dagger$. Particularly, it depends on $\mathbb{1}\{W_t=1\}$ and the neighborhood assignments. For example, if $W_t=1$ and $W_{t+1}=0$, then the period $t+1$ contributes to $0.5Y_t(1)$; similarly, every additional $W_{t'}=0$ for $t'>t$ contributes to $0.5Y_t(1)$. In this way, the weight of $Y_t(1)$ is given by

$$\sum_{k=1}^{T/2}0.5\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2}=0.25. \tag{49}$$

We also have the other direction with $t'<t$, so in expectation, we obtain $0.5Y_t(1)$ from $\widehat{\tau}^\dagger$. Applying the same argument for the control outcome, we get

$$\mathbb{E}[\widehat{\tau}^\dagger]=\frac{1}{2T}\sum_{t\in[T]}(Y_t(1)-Y_t(0))=\frac{1}{2}\tau. \tag{50}$$

D. Proof of Proposition 2

We follow the decomposition above, and given that both estimators are unbiased, the variance gap becomes

$$\mathrm{Var}(\widehat{\tau}^*)-\mathrm{Var}(\widehat{\tau})=\mathbb{E}[\widehat{\tau}\cdot\widehat{\tau}^\dagger]+\mathbb{E}[(\widehat{\tau}^\dagger)^2]-\frac{3}{4}\mathbb{E}[\widehat{\tau}^2]. \tag{51}$$

Note the variance of IPW estimator is well-known in the literature, so the key is to characterize $\mathbb{E}[\widehat{\tau}\cdot\widehat{\tau}^\dagger]$ and $\mathbb{E}[(\widehat{\tau}^\dagger)^2]$.

Variance from treatment only. We first check the terms related to the treatment. We start with $\mathbb{E}[\widehat{\tau}\cdot\widehat{\tau}^{\dagger}]$ and we want to expand this into the sum of products between potential outcomes. We have

$$\frac{2}{T^2}\mathbb{E}\left[\sum_{t\in[T]}\mathbb{1}\{W_t=1\}Y_t(1)\cdot\sum_{t\in[T]}\mathbb{1}\{W_t=0\}Y_t^{\dagger}(1)\right] \tag{52}$$

which is equivalent to

$$\frac{2}{T^2}\mathbb{E}\left[\sum_{t\in[T]}\mathbb{1}\{W_t=1\}Y_t(1)\cdot\sum_{t\in[T]}\mathbb{1}\{W_t=1\}\frac{Y_t(1)}{2}\sum_{t'\neq t}\mathbb{1}\{\boldsymbol{W}_{t':t}=\mathbf{0}\}\right]. \tag{53}$$

After normalized by T^2, to get the coefficient for $Y_t^2(1)$, we follow the same procedure for the proof above where we check the neighborhood assignments of period t. This is given by

$$2\sum_{k=1}^{T/2}\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2}=1. \tag{54}$$

To get the coefficient for $Y_t(1)Y_{t+1}(1)$, we see that the weight is non-zero only if both periods are in the treatment group, and then we check the assignments $t'>t+1$ and $t'<t$ as we did above. We have

$$2\sum_{k=1}^{T/2}\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2}=\frac{T/2}{T-1} \tag{55}$$

Following the similar way of analysis, we can generalize that the coefficient for $Y_t(1)Y_{t+q}(1)$ is

$$2\left(\sum_{k=1}^{T/2}\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2}+\sum_{k=1}^{q-1}\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2}\right). \tag{56}$$

The decomposition of the terms related to control outcomes is symmetric so we omit here.

Next we work on $\mathbb{E}[(\widehat{\tau}^{\dagger})^2]$. First, we analyze

$$\frac{1}{T^2}\mathbb{E}\left[\left(\sum_{t\in[T]}\mathbb{1}\{W_t=1\}\frac{Y_t(1)}{2}\sum_{t'\neq t}\mathbb{1}\{\boldsymbol{W}_{t':t}=\mathbf{0}\}\right)^2\right]. \tag{57}$$

For the coefficient for $Y_t^2(1)$, it can be written as

$$\frac{1}{4}\mathbb{E}\left[\mathbb{1}\{W_t=1\}\left(\sum_{t'\neq t}\mathbb{1}\{\boldsymbol{W}_{t':t}=\mathbf{0}\}\right)^2\right]=\frac{1}{4}\mathbb{E}\left[\sum_{t',t''\neq t}\mathbb{1}\{W_t=1,\boldsymbol{W}_{t':t}=\mathbf{0},\boldsymbol{W}_{t'':t}=\mathbf{0}\}\right] \tag{58}$$

For $t',t''<t$, we have

$$\frac{1}{4}\mathbb{E}\left[\sum_{t',t''\neq t}\mathbb{1}\{W_t=1,\boldsymbol{W}_{\min\{t',t''\}:t}=\mathbf{0}\}\right]=\frac{1}{4}\sum_{k=1}^{T/2}(2k-1)\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2} \tag{59}$$

The case of $t', t'' > t$ is symmetric. For $t' < t < t''$, we have

$$\frac{1}{4}\mathbb{E}\left[\sum_{t',t''\neq t} \mathbb{1}\{W_t = 1, \boldsymbol{W}_{t':t} = \boldsymbol{0}, \boldsymbol{W}_{t:t''} = \boldsymbol{0}\}\right] = \frac{1}{4}\sum_{k=2}^{T/2}(k-1)\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2} \tag{60}$$

Combining them together, we see that the coefficient is equivalent to

$$\frac{1}{2}\sum_{k=1}^{T/2}(3k-2)\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2}. \tag{61}$$

For the coefficient for $Y_t(1)Y_{t+1}(1)$, it can be written as

$$\frac{1}{2}\mathbb{E}\left[\left(\sum_{t'\neq t} \mathbb{1}\{W_t = 1, \boldsymbol{W}_{t':t} = \boldsymbol{0}\}\right)\left(\sum_{t''\neq t+1} \mathbb{1}\{W_{t+1} = 1, \boldsymbol{W}_{t'':t+1} = \boldsymbol{0}\}\right)\right] \tag{62}$$

The indicator is non-zero only if $t' < t, t'' > t+1$, so this becomes

$$\frac{1}{2}\sum_{k=1}^{T/2}(k-1)\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2} \tag{63}$$

For the coefficient for $Y_t(1)Y_{t+2}(1)$, it can be written as

$$\frac{1}{2}\mathbb{E}\left[\left(\sum_{t'\neq t} \mathbb{1}\{W_t = 1, \boldsymbol{W}_{t':t} = \boldsymbol{0}\}\right)\left(\sum_{t''\neq t+2} \mathbb{1}\{W_{t+2} = 1, \boldsymbol{W}_{t'':t+2} = \boldsymbol{0}\}\right)\right] \tag{64}$$

The part with $t' < t, t'' > t+2$ will be exactly the same as (63). The additional parts are $t' < t, t < t'' < t+2$, $t < t' < t+2, t'' > t+2$, and $t < t', t'' < t+2$, whose sum is given by

$$\sum_{k=2}^{T/2}\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2} + \frac{1}{2}\tbinom{T-3}{T/2-1}/\tbinom{T}{T/2} \tag{65}$$

Altogether, the coefficient is

$$\frac{1}{2}\sum_{k=2}^{T/2}(k-1)\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2} + \sum_{k=2}^{T/2}\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2} + \frac{1}{2}\tbinom{T-3}{T/2-1}/\tbinom{T}{T/2}. \tag{66}$$

In general $q > 2$, the coefficient for $Y_t(1)Y_{t+q}(1)$ is

$$\frac{1}{2}\sum_{k=2}^{T/2}(k-1)\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2} + \sum_{k=2}^{T/2}\min\{k-1, q-1\}\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2} + \frac{1}{2}\sum_{k=2}^{q-2}(k-1)\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2} + \frac{q^2+q-4}{4}\tbinom{T-1-q}{T/2-q+1}/\tbinom{T}{} \tag{67}$$

Finally, the normalized variance gap contains

$$\left(1 + \frac{1}{2}\sum_{k=1}^{T/2}(3k-2)\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2} - \frac{3}{2}\right)\sum_{t\in[T]} Y_t^2(1) = \frac{1}{T^2}\left(\frac{1}{2}\sum_{k=1}^{T/2}(3k-2)\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2} - \frac{1}{2}\right)\sum_{t\in[T]} Y_t^2(1). \tag{68}$$

As $T \to +\infty$, the coefficient converges to $\beta_0^+ = 1/2 - 3/T$. Similarly, we can get the coefficient $\beta_1^+ = -7/8 + 15/(8T)$. In general $q \geq 2$, we have

$$2\left(\sum_{k=1}^{T/2}\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2} + \sum_{k=1}^{q-1}\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2}\right) + \frac{1}{2}\sum_{k=2}^{T/2}(k-1)\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2} + \sum_{k=2}^{q}(k-1)\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2}$$
$$+ \sum_{k=q+1}^{T/2}(q-1)\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2} + \frac{1}{2}\sum_{k=2}^{q-2}(k-1)\tbinom{T-2-k}{T/2-k}/\tbinom{T}{T/2} + \frac{q^2+q-4}{4}\tbinom{T-1-q}{T/2-q+1}/\tbinom{T}{T/2} - \frac{3(T-2)}{2(T-1)}. \tag{69}$$

The first two terms converge to $1 - 2^{-q} + \frac{1}{T}(1 + \frac{q^2-3q}{2^{q+1}})$; the third term converges to $\frac{1}{8} - \frac{1}{8T}$; the fourth term converges to $\frac{1}{4} - \frac{q+1}{2^{q+2}} + \frac{1}{T}(\sum_{k=2}^{q}(k-1)\,(-k^2+5k-2)\,2^{-k-3})$; the fifth term converges to $\frac{q-1}{2^{q+2}} - \frac{1}{T}\frac{(q-1)(q+1)(q-2)}{2^{q+3}}$; the sixth term converges to $\frac{1}{8} - \frac{q-1}{2^{q+1}} + \frac{1}{T}(\sum_{k=2}^{q-2}(k-1)\frac{-k^2+5k-2}{2^{k+4}})$; the seventh term converges to $\frac{q^2+q-4}{2^{q+3}} + \frac{\left(q^2+q-4\right)\left(-q^2+7q-8\right)}{2^{q+4}}\frac{1}{T}$; the last terms converges to $\frac{3}{2} - \frac{3}{2T}$. This can be further aggregated and reformulated as

$$\beta_q^+ = \frac{q^2-3q-12}{2^{q+3}} + \frac{f(q)}{T} \tag{70}$$

where $f(q)$ denotes the weight for $\frac{1}{T}$ aggregated from the terms above.

Variance from treatment and control. We then check the cross-terms related to the treatment and control. We also start with $\mathbb{E}[\widehat{\tau} \cdot \widehat{\tau}^\dagger]$. That is,

$$-\frac{2}{T^2}\mathbb{E}\left[\sum_{t\in[T]}\mathbb{1}\{W_t = 1\}Y_t(1) \cdot \sum_{t\in[T]}\mathbb{1}\{W_t = 1\}Y_t^\dagger(0)\right] \tag{71}$$

which is equivalent to

$$-\frac{2}{T^2}\mathbb{E}\left[\sum_{t\in[T]}\mathbb{1}\{W_t = 1\}Y_t(1) \cdot \sum_{t\in[T]}\mathbb{1}\{W_t = 0\}\frac{Y_t(0)}{2}\sum_{t'\neq t}\mathbb{1}\{\boldsymbol{W}_{t':t} = \mathbf{1}\}\right]. \tag{72}$$

It is easy to see that the coefficient for $Y_t(1)Y_t(0)$ is zero. After normalized by T^2, the coefficient for $Y_t(1)Y_{t+1}(0)$ can be written as

$$-2\mathbb{E}\left[\mathbb{1}\{\boldsymbol{W}_{t:t+1} = (1,0)\}\left(\sum_{t'\neq t+1}\mathbb{1}\{\boldsymbol{W}_{t':t+1} = \mathbf{1}\}\right)\right] \tag{73}$$

which is equivalent to

$$-2\left(2\sum_{k=2}^{T/2}\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2} + \tbinom{T-2}{T/2-1}/\tbinom{T}{T/2}\right) = -\frac{3T-4}{2(T-1)} \tag{74}$$

In general, the coefficient for $Y_t(1)Y_{t+q}(0)$ is

$$-2\mathbb{E}\left[\mathbb{1}\{W_t = 1, W_{t+q} = 0\}\left(\sum_{t'\neq t+1}\mathbb{1}\{\boldsymbol{W}_{t':t+q} = \mathbf{1}\}\right)\right] \tag{75}$$

which is equivalent to

$$-2\left(\sum_{k=2}^{T/2}\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2}+\sum_{k=1}^{q-1}\tbinom{T-2-k}{T/2-k-1}/\tbinom{T}{T/2}+\sum_{k=q}^{T/2}\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2}\right). \tag{76}$$

The analysis of $Y_t(0)Y_{t+q}(1)$ is symmetric so we omit here.

Next we work on $\mathbb{E}[(\widehat{\tau}^\dagger)^2]$. First, we analyze

$$-\frac{2}{T^2}\mathbb{E}\left[\sum_{t\in[T]}\mathbb{1}\{W_t=0\}Y_t^\dagger(1)\cdot\sum_{t\in[T]}\mathbb{1}\{W_t=1\}Y_t^\dagger(0)\right] \tag{77}$$

which is equivalent to

$$-\frac{2}{T^2}\mathbb{E}\left[\sum_{t\in[T]}\mathbb{1}\{W_t=1\}\frac{Y_t(1)}{2}\sum_{t'\neq t}\mathbb{1}\{\boldsymbol{W}_{t':t}=\mathbf{0}\}\cdot\sum_{t\in[T]}\mathbb{1}\{W_t=0\}\frac{Y_t(0)}{2}\sum_{t'\neq t}\mathbb{1}\{\boldsymbol{W}_{t':t}=\mathbf{1}\}\right]. \tag{78}$$

It is easy to see that the coefficient for $Y_t(0)Y_t(1)$ is zero. For the coefficient for $Y_t(1)Y_{t+1}(0)$, it can be written as

$$-\frac{1}{2}\mathbb{E}\left[\mathbb{1}\{W_t=1\}\sum_{t'\neq t}\mathbb{1}\{\boldsymbol{W}_{t':t}=\mathbf{0}\}\cdot\mathbb{1}\{W_{t+1}=0\}\sum_{t''\neq t+1}\mathbb{1}\{\boldsymbol{W}_{t'':t+1}=\mathbf{1}\}\right] \tag{79}$$

For $t'<t, t''>t+1$, we have

$$-\frac{1}{2}\sum_{k=2}^{T/2}\sum_{l=2}^{T/2}\tbinom{T-k-l}{T/2-k}/\tbinom{T}{T/2}. \tag{80}$$

For $t'>t, t''<t+1$, we have

$$-\frac{1}{2}\sum_{k=1}^{T/2}\sum_{l=1}^{T/2}\tbinom{T-k-l}{T/2-k}/\tbinom{T}{T/2}. \tag{81}$$

We have non-zero terms for $t'<t, t''=t$ and $t''>t+1, t'=t+1$. Their sum is

$$-\sum_{k=2}^{T/2}\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2}. \tag{82}$$

Putting together, the coefficient is

$$-\frac{1}{2}\left(2\sum_{k=1}^{T/2}\sum_{l=1}^{T/2}\tbinom{T-k-l}{T/2-k}/\tbinom{T}{T/2}-\tbinom{T-2}{T/2-1}/\tbinom{T}{T/2}\right), \tag{83}$$

In general $q\geq 2$, the coefficient for $Y_t(1)Y_{t+q}(0)$ is

$$-\frac{1}{2}\left(\sum_{k=2}^{T/2}\sum_{l=2}^{T/2}\tbinom{T-k-l}{T/2-k}/\tbinom{T}{T/2}+\sum_{k=2}^{q-1}\sum_{l=2}^{q+1-k}\tbinom{T-k-l}{T/2-k}/\tbinom{T}{T/2}\right)-\sum_{k=2}^{T/2}\sum_{l=2}^{q}\tbinom{T-k-l}{T/2-k}/\tbinom{T}{T/2}-\sum_{k=2}^{T/2}\tbinom{T-q-k}{T/2-k}/\tbinom{T}{T/2}. \tag{84}$$

Finally, the variance gap contains

$$-\frac{1}{T^2}\left(\frac{1}{2}\left(\sum_{k=2}^{T/2}\sum_{l=2}^{T/2}\tbinom{T-k-l}{T/2-k}/\tbinom{T}{T/2}+\sum_{k=1}^{T/2}\sum_{l=1}^{T/2}\tbinom{T-k-l}{T/2-k}/\tbinom{T}{T/2}+2\sum_{k=2}^{T/2}\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2}\right)-\frac{2}{T-1}\right)\sum_{t\in[T]}Y_t(1)Y_{t+1}(0). \tag{85}$$

As $T\to+\infty$, the coefficient converges to $-\frac{7}{8}+\frac{17}{8T}$. In general $q\geq 2$, we have

$$\begin{aligned}&-2\left(\sum_{k=2}^{T/2}\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2}+\sum_{k=1}^{q-1}\tbinom{T-2-k}{T/2-k-1}/\tbinom{T}{T/2}+\sum_{k=q}^{T/2}\tbinom{T-1-k}{T/2-k}/\tbinom{T}{T/2}\right)\\&-\frac{1}{2}\left(\sum_{k=2}^{T/2}\sum_{l=2}^{T/2}\tbinom{T-k-l}{T/2-k}/\tbinom{T}{T/2}+\sum_{k=2}^{q-1}\sum_{l=2}^{q+1-k}\tbinom{T-k-l}{T/2-k}/\tbinom{T}{T/2}\right)-\sum_{k=2}^{T/2}\sum_{l=2}^{q}\tbinom{T-k-l}{T/2-k}/\tbinom{T}{T/2}-\sum_{k=2}^{T/2}\tbinom{T-q-k}{T/2-k}/\tbinom{T}{T/2}+\frac{3T}{2(T-1)}\end{aligned} \tag{86}$$

The first line together converges to $-1-\frac{1}{2^q}+\frac{1}{T}(1+o(1))$; the second line together converges to $1+\frac{q}{2^{q+2}}+\frac{1}{T}(1+o(1))$. This can be further reformulated as

$$\beta_q^- = \frac{q-4}{2^{q+2}}+O(1/T). \tag{87}$$

E. Proof of Corollary 1

We first work on the cross moments for $Y_t(1)Y_{t+q}(1)$, which can be represented as

$$\sum_{q=0}^{T/2}\beta_q^+\sum_{t\in[T]}Y_t(1)Y_{t+q}(1). \tag{88}$$

We aim to show this is equivalent to

$$\sum_{q=0}^{T/2}\beta_q^+\sum_{t\in[T]}(Y_t(1)-\bar{Y}(1))(Y_{t+q}(1)-\bar{Y}(1)). \tag{89}$$

By expanding both expressions, we realize that it suffices to show that

$$\sum_{q=0}^{T/2-1}\beta_q^+ + \frac{1}{2}\beta_{T/2}^+ = 0. \tag{90}$$

By plugging the value of β_q^+ derived before, as $T\to\infty$, we have

$$\sum_{q=0}^{T/2-1}\beta_q^+ + \frac{1}{2}\beta_{T/2}^+ = \frac{1}{2}+\sum_{q=1}^{T/2-1}\left(\frac{q^2-3q-12}{2^{q+3}}\right)+\frac{1}{T}\left(-3+\frac{15}{8}+\sum_{q=2}^{T/2-1}f(q)\right)+\frac{1}{2}\beta_{T/2}^+\to 0. \tag{91}$$

Similarly, we can also check that

$$\sum_{q=0}^{T/2-1}\beta_q^- + \frac{1}{2}\beta_{T/2}^- = 0. \tag{92}$$

This leads to

$$\sum_{q=0}^{T/2}\beta_q^-\sum_{t\in[T]}Y_t(1)Y_{t+q}(0) = \sum_{q=0}^{T/2}\beta_q^-\sum_{t\in[T]}(Y_t(1)-\bar{Y}(1))(Y_{t+q}(0)-\bar{Y}(0)). \tag{93}$$

Finally, based on the following transformation for any t and $t+q$:

$$\begin{aligned}&(Y_t(1)-\bar{Y}(1))(Y_{t+q}(0)-\bar{Y}(0))+(Y_t(0)-\bar{Y}(0))(Y_{t+q}(1)-\bar{Y}(1))\\&=(Y_t(1)-\bar{Y}(1))(Y_{t+q}(1)-\bar{Y}(1))+(Y_t(0)-\bar{Y}(0))(Y_{t+q}(0)-\bar{Y}(0))-(\tau_t-\tau)(\tau_{t+q}-\tau)\end{aligned} \tag{94}$$

and normalized by $\mathrm{Var}(Y_t(w))$ and $\mathrm{Var}(\tau_t)$, we end up with

$$\begin{aligned}&T\left(\mathrm{Var}(\widehat{\tau}^*)-\mathrm{Var}(\widehat{\tau})\right)\\&=\mathrm{Var}(Y_t(1))\left(\sum_{q=0}^{T/2}(\beta_q^+ +\beta_q^-)\rho_q(1)\right)+\mathrm{Var}(Y_t(0))\left(\sum_{q=0}^{T/2}(\beta_q^+ +\beta_q^-)\rho_q(0)\right)-\mathrm{Var}(\tau_t)\left(\sum_{q=0}^{T/2}\beta_q^-\rho_q^\tau\right).\end{aligned} \tag{95}$$

F. Proof of Error Probability

To show that the first-stage decision from the robust optimization formulation 12 has a 50% chance of selecting the treatment with no effect, we need to show the following proposition.

PROPOSITION 3. *Under the null $\tau=0$ and the condition that $\max_t|Y_t(w)|\le M<\infty$ for all T, the error rate satisfies that as $T\to\infty$,*

$$\mathbb{P}(\widehat{\tau}^*>0)\to 0.5. \tag{96}$$

To prove it, we first recall the decomposition

$$\widehat{\tau}^*=\widehat{\tau}+\widehat{\tau}^\dagger-\frac{1}{2}\widehat{\tau}. \tag{97}$$

Specifically, we can represent the second part as a weighted average of the potential outcomes:

$$\widehat{\tau}^\dagger-\frac{1}{2}\widehat{\tau}=\frac{1}{T}\sum_{t=1}^{T}\left[\left(\frac{1}{2}B_t(1)-1\right)W_tY_t(1)-\left(\frac{1}{2}B_t(0)-1\right)(1-W_t)Y_t(0)\right] \tag{98}$$

where $B_t(w)$ denotes the number of times the outcome $Y_t(w)$ is "borrowed" to impute the outcome at other periods. We also denote the second part by $d(\boldsymbol{W})$, showing it is the difference between $\widehat{\tau}^*$ and $\widehat{\tau}$ and is dependent of the assignment path $\boldsymbol{W}$. For $t\in\{1,\dots,T\}$ let $\boldsymbol{W}^{(t)}$ be $\boldsymbol{W}$ with the t-th period assignment flipped. Define the (random) influence numbers

$$\delta_t(\boldsymbol{W}):=\left|\sqrt{T}(d(\boldsymbol{W})-d(\boldsymbol{W}^{(t)}))\right| \tag{99}$$

The change at t affects not only the outcome at period t, but also the outcomes at other periods that used to borrow from t and now need to borrow from a new period. Let the maximal length of a consecutive run of identical labels in $\boldsymbol{W}$ be

$$B_{\max}(\boldsymbol{W}):=\max\{\text{ longest consecutive 0 's, longest consecutive 1 's }\}.$$

If there are ℓ periods in control on the left of t, and ℓ' periods in control on the right of t, we can see that

$$\delta_t(\boldsymbol{W}) \leq \frac{\sqrt{T}}{T}\left[2M + M\left(\ell + \ell'\right)\right] \leq \frac{2M}{\sqrt{T}}\left(1 + B_{\max}(\boldsymbol{W})\right). \tag{100}$$

Under the balanced complete randomization design, the classical Erdős–Rényi bound gives $B_{\max}(\boldsymbol{W}) = O_p(\log T)$, which means $\delta_t(\boldsymbol{W}) = O_p(\log T/\sqrt{T})$. It is well known that the IPW estimator $\widehat{\tau}$ shares the single-period influence in $O(1/\sqrt{T})$, so combining them together, the multivariate Berry-Esseen condition is satisfied. Rinott and Rotar (1997) give the joint normal limit of $\sqrt{T}(\widehat{\tau}, \widehat{\tau}^{\dagger} - \frac{1}{2}\widehat{\tau})$ that is centered around 0. Finally, the error rate convergence follows directly.